



Calibration Cert# 2166.01

FLUKE®

Certificate of Calibration

Everett Service Center

Certificate Number: EVL918262

Data Type: As-Left
Result Summary: Measurement Results < Limits
Manufacturer: Fluke
Model: 5522A
Serial Number: 2806902
Description: Calibrator
Option(s): PQ/SC600

Calibration Date: 04-Oct-2023
Calibration Due: 04-Oct-2025
Certificate Date: 05-Oct-2023
Temperature: 22.9 °C
Humidity: 51.8 %

Procedure: Fluke 5522A:(1 year) VER Combine **Revision:** 5.4

Customer: MODERN SURVEYING CALIBRATION & TESTING LABS(FZE)
City: ERBIL **Country:** IQ

Purchase Order: PO1230028 **RMA:** 32677155
Asset ID: MSL/E/001

This calibration is traceable to the International System of Units (SI), through National Metrology Institutes (NIST, PTB, NRC, NPL, etc.), radiometric techniques, or natural physical constants. This certificate applies only to the item identified and shall not be reproduced other than in full, without the specific written approval by Fluke Corporation. Calibration certificates without signature are not valid. The calibration has been completed in accordance with Fluke Electronics Corporation Quality System Document 111.0 Revision 125 and/or Fluke 17025 Quality Manual QSD 111.41 Revision 008.

The Data Type found in this certificate must be interpreted as:

- As - Found Calibration data collected before the unit is adjusted and / or repaired.
- As - Left Calibration data collected after the unit has been adjusted and / or repaired.
- Found-Left Calibration data collected without any adjustment and / or repair performed.

This calibration conforms to the requirements of ISO/IEC 17025:2017 and ANSI/NCSL Z540-1-1994 (R2002).

In the attached measurement results, deviation may be expressed with units, Measured Value (MV) - Nominal Value (NV) or as a proportion of the nominal value ((MV-NV)/NV), expressed without units with a scalar multiplier such as % (0.01), or as a ratio of the units (mA/A, μ V/V, etc.) Descriptions such as μ A/A, μ V/V, and others, where used to annotate results or column headings are the preferred replacements for what was historically labeled as "ppm" or parts-per-million and described the results in that column, unless otherwise noted by units symbols.

Where applicable, the expanded uncertainty of measurement at the time of test is given in the following pages. They are calculated in accordance with the method described in the ISO Guide to the Expression of Uncertainty in Measurement (GUM). The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k, such that the confidence level approximates 95%.

This calibration certificate may contain data that is not covered by the A2LA Scope of Accreditation. Unaccredited material, where applicable is indicated by an asterisk (*), or confined to clearly marked sections. Functional (Pass / Fail) tests are not accredited.

No statement of compliance with specifications is made or implied on this certificate. However, measurement results are reviewed, where applicable, to establish where any measurement result exceeded the manufacturer's specifications.

Measurement results greater than limits of error are indicated by '!'.
Measurement results less than limits of error are indicated by '!'.



Accredited

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Cert #: EVL918262
Cal Date: 04-Oct-2023
Due Date: 04-Oct-2025
S/N : 2806902
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Cert #: EVL918262
Date: 04-Oct-2023
Due: 04-Oct-2025

Jeff Card
Calibration Technician

Certificate Number: EVL918262

Date of Calibration: 04-Oct-2023

Standards Used

Asset	Description	Cal-Date	Cal-Due
J772	Agilent 3458A Multimeter	08-Dec-2022	08-Dec-2023
18199	Agilent E4418B EPM Series Power Meter	06-Oct-2022	06-Oct-2023
15923	Clarke-Hess 6000A Phase Meter	03-Jun-2023	03-Jun-2024
18054	Fluke 1504 Thermometer Readout	22-Dec-2022	22-Dec-2023
7678	Fluke 5450A Resistance Calibrator	19-Jun-2023	19-Dec-2023
9978	Fluke 5450A Resistance Calibrator	02-Feb-2023	02-Feb-2024
13676	Fluke 5500A Calibrator	12-Mar-2023	12-Mar-2024
18069	Fluke 5640 Thermistor Probe	28-Nov-2022	28-Nov-2023
10073	Fluke 5790A AC Measurement Standard	13-Jul-2023	13-Oct-2023
9911	Fluke 5790A AC Measurement Standard	02-Dec-2022	02-Dec-2023
9911S	Fluke 5790A/A40 Set Calibrator Characterization	25-Sep-2023	25-Oct-2023
9106	Fluke 742A-1 Resistance Standard	20-Apr-2023	20-Apr-2024
9473	Fluke 742A-10 Resistance Standard	20-Apr-2023	20-Apr-2024
9019	Fluke 742A-100 Resistance Standard	20-Apr-2023	20-Apr-2024
9036	Fluke 742A-1k Resistance Standard	20-Apr-2023	20-Apr-2024
5823	Fluke 752A Reference Divider	01-Mar-2023	01-Mar-2024
16053	Fluke 8846A Precision Multimeter	15-Feb-2023	15-Feb-2024
10835	Fluke 910R GPS Frequency Standard	13-Sep-2023	13-Sep-2024
9866	Fluke PM6304C Programmable Automatic RCL Meter	24-Feb-2023	24-Feb-2024
9862	FLUKE PM6680B Programmable Timer/Counter	22-Jun-2023	22-Jun-2024
J1217	FLUKE PM6680B Programmable Timer/Counter	17-Mar-2023	17-Mar-2024
11899	Fluke SC Cap Set Oscilloscope Capacitance Artifact	05-Oct-2022	05-Oct-2023
12083	Fluke SC RES SET Oscilloscope Resistance Artifact	22-Sep-2023	22-Dec-2023
15980	Fluke SL984 Resistance Standard	15-Jun-2023	15-Jun-2024
10384	Fluke Y5020 Current Shunt	13-Jul-2023	13-Jan-2024
8976	Fluke Y5020 Current Shunt	03-May-2023	03-Feb-2024
10066	Hewlett Packard 3458A Multimeter	17-Feb-2023	17-Nov-2023
13629	Hewlett Packard 3458A Multimeter	23-Jan-2023	23-Jan-2024
16847	Hewlett Packard 8481D Power Sensor	29-Dec-2022	29-Dec-2023
16736	Hewlett Packard 8482A Power Sensor	06-Mar-2023	06-Mar-2024
13665	Hewlett Packard E4408B Spectrum Analyzer	27-Feb-2023	27-Feb-2024
18064	Krohn Hite 6900B Distortion Analyzer	15-Jun-2023	15-Jun-2024
18344	Omega TJ36-ICIN-18U-12-SB-SMPW-M "J" Type Rugged Transition Joint Probe	01-Jun-2023	01-Jun-2024
16898	Tektronix 11801B Digital Sampling Oscilloscope	15-Oct-2022	15-Oct-2023
18193	Tektronix FCA3103 Timer/Counter	25-Sep-2023	25-Sep-2024
16899C	Tektronix SD-22 Rise Sampling Head	15-Oct-2022	15-Oct-2023

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
1 Year Specification Limits shown					
DC VOLTAGE ACCURACY TEST (NORMAL)					
330 mV Range					
0.0000 mV	0.0000	-0.00013	-0.00100	0.00100	2.4e-007 V
329.000 mV	329.000	329.0000	328.9924	329.0076	1.4e-006 V
-329.000 mV	-329.000	-328.9991	-329.0076	-328.9924	1.4e-006 V
3.3 V Range					
0.000000 V	0.000000	0.0000005	-0.0000020	0.0000020	5.0e-007 V
1.000000 V	1.000000	1.0000005	0.9999870	1.0000130	3.5e-006 V
-1.000000 V	-1.000000	-1.0000006	-1.0000130	-0.9999870	3.5e-006 V
3.290000 V	3.290000	3.2900041	3.2899618	3.2900382	9.3e-006 V
-3.290000 V	-3.290000	-3.2900026	-3.2900382	-3.2899618	9.3e-006 V
33 V Range					
0.00000 V	0.00000	-0.000002	-0.000020	0.000020	4.6e-006 V
10.00000 V	10.00000	10.000014	9.999860	10.000140	2.7e-005 V
-10.00000 V	-10.00000	-10.000023	-10.000140	-9.999860	2.8e-005 V
32.90000 V	32.90000	32.900025	32.899585	32.900415	1.0e-004 V
-32.90000 V	-32.90000	-32.900035	-32.900415	-32.899585	1.0e-004 V
330 V Range					
50.0000 V	50.0000	50.00017	49.99895	50.00105	1.5e-004 V
-50.0000 V	-50.0000	-50.00017	-50.00105	-49.99895	1.5e-004 V
329.0000 V	329.0000	329.00017	328.99393	329.00607	9.9e-004 V
-329.0000 V	-329.0000	-329.00016	-329.00607	-328.99393	9.9e-004 V
1000 V Range					
334.000 V	334.000	334.0009	333.9925	334.0075	1.1e-003 V
-334.000 V	-334.000	-334.0017	-334.0075	-333.9925	1.0e-003 V
900.000 V	900.000	900.0013	899.9823	900.0177	2.6e-003 V
-900.000 V	-900.000	-900.0026	-900.0177	-899.9823	2.7e-003 V
1020.000 V	1020.000	1020.0006	1019.9801	1020.0199	3.0e-003 V
-1020.000 V	-1020.000	-1020.0026	-1020.0199	-1019.9801	3.0e-003 V
DC VOLTAGE ACCURACY TEST (AUX)					
330 mV Range					
0.000 mV	0.000	0.0012	-0.3500	0.3500	4.1e-006 V
329.000 mV	329.000	328.9992	328.5184	329.4816	9.9e-006 V
-329.000 mV	-329.000	-328.9961	-329.4816	-328.5184	9.9e-006 V
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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
3.3 V Range					
0.33000 V	0.33000	0.329998	0.329518	0.330482	5.0e-006 V
-0.33000 V	-0.33000	-0.330001	-0.330482	-0.329518	5.0e-006 V
3.29000 V	3.29000	3.290002	3.288334	3.291666	3.3e-005 V
-3.29000 V	-3.29000	-3.289999	-3.291666	-3.288334	3.3e-005 V
7.0000 V	7.0000	7.00000	6.99685	7.00315	2.1e-004 V
-7.0000 V	-7.0000	-7.00002	-7.00315	-6.99685	2.1e-004 V
DC OFFSET ACCURACY TEST					
50 mV Range					
0.000 mV	0.000	-0.0030	-0.0330	0.0330	2.8e-006 V
50.000 mV	50.000	49.9560	49.9170	50.0830	5.0e-006 V
-50.000 mV	-50.000	-49.9710	-50.0830	-49.9170	5.0e-006 V
500 mV Range					
0.00 mV	0.00	-0.000	-0.330	0.330	4.4e-005 V
500.00 mV	500.00	499.946	499.170	500.830	6.2e-005 V
-500.00 mV	-500.00	-500.039	-500.830	-499.170	6.5e-005 V
5 V Range					
0.0000 V	0.0000	0.00016	-0.00330	0.00330	8.5e-004 V
5.0000 V	5.0000	4.99967	4.99170	5.00830	8.2e-004 V
-5.0000 V	-5.0000	-5.00034	-5.00830	-4.99170	1.1e-003 V
50 V Range					
0.0000 V	0.0000	0.01644	-0.03300	0.03300	3.4e-003 V
50.0000 V	50.0000	50.00783	49.91700	50.08300	4.0e-003 V
-50.0000 V	-50.0000	-49.96415	-50.08300	-49.91700	5.0e-003 V
DC CURRENT ACCURACY TEST					
330 µA Range					
0.00000 µA	0.00000	0.000266	-0.020000	0.020000	2.2e-009 A
33.000 µA	33.000	33.0006	32.9750	33.0250	4.5e-010 A
-33.000 µA	-33.000	-32.9999	-33.0250	-32.9750	4.5e-010 A
190.000 µA	190.000	190.0025	189.9515	190.0485	3.0e-009 A
-190.000 µA	-190.000	-190.0012	-190.0485	-189.9515	3.0e-009 A
329.000 µA	329.000	329.0040	328.9306	329.0694	3.1e-009 A
-329.000 µA	-329.000	-329.0025	-329.0694	-328.9306	3.1e-009 A
3.3 mA Range					
0.00000 mA	0.00000	-0.000001	-0.000050	0.000050	1.6e-008 A
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Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
1.90000 mA	1.90000	1.900017	1.899760	1.900240	2.3e-008 A
-1.90000 mA	-1.90000	-1.900014	-1.900240	-1.899760	2.3e-008 A
3.29000 mA	3.29000	3.290029	3.289621	3.290379	3.1e-008 A
-3.29000 mA	-3.29000	-3.290026	-3.290379	-3.289621	3.1e-008 A
33 mA Range					
0.0000 mA	0.0000	-0.00002	-0.00025	0.00025	1.1e-007 A
19.0000 mA	19.0000	19.00011	18.99785	19.00215	2.3e-007 A
-19.0000 mA	-19.0000	-19.00013	-19.00215	-18.99785	2.3e-007 A
32.9000 mA	32.9000	32.90024	32.89646	32.90354	3.1e-007 A
-32.9000 mA	-32.9000	-32.90027	-32.90354	-32.89646	3.1e-007 A
330 mA Range					
0.0000 mA	0.0000	-0.00006	-0.00250	0.00250	2.8e-007 A
190.000 mA	190.000	190.0017	189.9785	190.0215	3.6e-006 A
-190.000 mA	-190.000	-190.0016	-190.0215	-189.9785	3.6e-006 A
329.000 mA	329.000	329.0046	328.9646	329.0354	6.6e-006 A
-329.000 mA	-329.000	-329.0069	-329.0354	-328.9646	6.6e-006 A
3 A Range					
0.00000 A	0.00000	0.000000	-0.000040	0.000040	2.7e-006 A
1.09000 A	1.09000	1.089997	1.089742	1.090258	2.2e-005 A
-1.09000 A	-1.09000	-1.090033	-1.090258	-1.089742	2.2e-005 A
2.99000 A	2.99000	2.989865	2.988824	2.991176	1.9e-004 A
-2.99000 A	-2.99000	-2.989832	-2.991176	-2.988824	1.9e-004 A
20 A Range					
0.0000 A	0.0000	0.00000	-0.00050	0.00050	1.2e-004 A
10.9000 A	10.9000	10.90015	10.89405	10.90595	6.5e-004 A
-10.9000 A	-10.9000	-10.90050	-10.90595	-10.89405	6.5e-004 A
20.0000 A	20.0000	20.00497	19.97925	20.02075	1.8e-003 A
-20.0000 A	-20.0000	-20.00686	-20.02075	-19.97925	1.8e-003 A

PHASE ACCURACY TEST (Voltage vs. Voltage)
3 V, 3V

0.00 ° @ 65 Hz	0.00	-0.014	-0.100	0.100	5.0e-002 °
0.00 ° @ 400 Hz	0.00	-0.003	-0.250	0.250	5.3e-002 °
0.00 ° @ 1 kHz	0.00	-0.007	-0.500	0.500	5.2e-002 °
0.00 ° @ 5 kHz	0.00	0.078	-2.500	2.500	2.0e-001 °
0.00 ° @ 10 kHz	0.00	0.149	-5.000	5.000	3.8e-001 °

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.00 ° @ 30 kHz	0.00	0.706	-10.000	10.000	5.0e-001 °
60.00 ° @ 65 Hz	60.00	59.986	59.900	60.100	5.0e-002 °
60.00 ° @ 400 Hz	60.00	59.992	59.750	60.250	5.3e-002 °
60.00 ° @ 1 kHz	60.00	59.999	59.500	60.500	5.2e-002 °
60.00 ° @ 5 kHz	60.00	60.136	57.500	62.500	2.0e-001 °
60.00 ° @ 10 kHz	60.00	60.391	55.000	65.000	3.8e-001 °
60.00 ° @ 30 kHz	60.00	60.562	50.000	70.000	5.0e-001 °
90.00 ° @ 65 Hz	90.00	89.982	89.900	90.100	5.0e-002 °
90.00 ° @ 400 Hz	90.00	89.998	89.750	90.250	5.3e-002 °
90.00 ° @ 1 kHz	90.00	89.988	89.500	90.500	5.2e-002 °
90.00 ° @ 5 kHz	90.00	90.080	87.500	92.500	2.0e-001 °
90.00 ° @ 10 kHz	90.00	90.142	85.000	95.000	3.8e-001 °
90.00 ° @ 30 kHz	90.00	90.814	80.000	100.000	5.0e-001 °

30 V, 3V

90.00 ° @ 65 Hz	90.00	89.970	89.900	90.100	5.0e-002 °
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50 V, 3V

90.00 ° @ 65 Hz	90.00	89.973	89.900	90.100	5.0e-002 °
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PHASE ACCURACY TEST (Voltage vs. Current)

0° (30 mV, 300mA @ 65 Hz)	0.00	-0.007	-0.100	0.100	5.0e-002 °
0° (30 mV, 300mA @ 1 kHz)	0.00	0.097	-0.500	0.500	5.2e-002 °
0° (30 mV, 300mA @ 30 kHz)	0.00	2.573	-10.000	10.000	5.0e-001 °
0° (200 mV, 2 A @ 65 Hz)	0.00	-0.001	-0.100	0.100	5.0e-002 °
0° (50 mV, 5 A @ 65 Hz)	0.00	-0.009	-0.100	0.100	5.0e-002 °
0° (50 mV, 5 A @ 400 Hz)	0.00	-0.070	-0.250	0.250	9.5e-002 °
0° (3.3 V, 300 mA @ 65 Hz)	0.00	-0.008	-0.100	0.100	5.0e-002 °
0° (3.3 V, 2 A @ 65 Hz)	0.00	0.009	-0.100	0.100	5.0e-002 °
0° (3.3 V, 5 A @ 65 Hz)	0.00	-0.031	-0.100	0.100	5.0e-002 °
0° (3.3 V, 5 A @ 400 Hz)	0.00	-0.075	-0.250	0.250	9.5e-002 °
0° (33 V, 300 mA @ 65 Hz)	0.00	-0.018	-0.100	0.100	5.0e-002 °
0° (33 V, 2 A @ 65 Hz)	0.00	-0.014	-0.100	0.100	5.0e-002 °
0° (33 V, 5 A @ 65 Hz)	0.00	-0.048	-0.100	0.100	5.0e-002 °
0° (33 V, 5 A @ 400 Hz)	0.00	-0.039	-0.250	0.250	9.5e-002 °
60° (30 mV, 300mA @ 65 Hz)	60.00	60.002	59.900	60.100	5.0e-002 °
60° (200 mV, 2 A @ 65 Hz)	60.00	60.007	59.900	60.100	5.0e-002 °
90° (3.3 V, 300 mA @ 65 Hz)	90.00	90.010	89.900	90.100	5.0e-002 °

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			Lower Limit	Upper Limit	
90° (3.3 V, 2 A @ 65 Hz)	90.00	90.018	89.900	90.100	5.0e-002 °
90° (33 V, 300 mA @ 65 Hz)	90.00	89.987	89.900	90.100	5.0e-002 °
90° (33 V, 2 A @ 65 Hz)	90.00	89.991	89.900	90.100	5.0e-002 °
60° (200 mV, 20 A @ 65 Hz)	60.00	59.992	59.900	60.100	5.0e-002 °
60° (200 mV, 20 A @ 400 Hz)	60.00	59.908	59.750	60.250	9.5e-002 °
90° (3.3 V, 20 A @ 65 Hz)	90.00	90.002	89.900	90.100	5.0e-002 °
90° (3.3 V, 20 A @ 400 Hz)	90.00	89.899	89.750	90.250	9.5e-002 °
90° (33 V, 20 A @ 65 Hz)	90.00	89.975	89.900	90.100	5.0e-002 °
90° (33 V, 20 A @ 400 Hz)	90.00	89.920	89.750	90.250	9.5e-002 °

FREQUENCY ACCURACY TEST

119.0 Hz @ 3 V	119.00000	118.999960	118.999697	119.000303	3.9e-005 Hz
120.0 Hz @ 3 V	120.00000	119.999964	119.999695	120.000305	6.4e-005 Hz
1.00 kHz @ 3 V	1.000000	0.9999997	0.9999975	1.0000025	1.2e-004 Hz
100 kHz @ 3 V	100.0000	99.99997	99.99975	100.00025	1.1e-002 Hz
500 kHz @ 3 V	500.0000	499.99985	499.99875	500.00125	4.5e-002 Hz

DISTORTION AND NOISE TEST (NORMAL)

300 mV Range

0.000 % @ 10 Hz	0.000	0.0491	0.0000	0.1800	*
0.000 % @ 45 Hz	0.000	0.0109	0.0000	0.0650	*
0.000 % @ 10 kHz	0.000	0.0109	0.0000	0.0650	*
0.000 % @ 20 kHz	0.000	0.0122	0.0000	0.0900	*
0.000 % @ 50 kHz	0.000	0.0260	0.0000	0.1800	*
0.000 % @ 100 kHz	0.000	0.0441	0.0000	0.2300	*
0.000 % @ 500 kHz	0.000	0.0934	0.0000	0.9300	*

300 V Range

0.000 % @ 45 Hz	0.000	0.0101	0.0000	0.1583	*
0.000 % @ 1 kHz	0.000	0.0118	0.0000	0.0583	*
0.000 % @ 10 kHz	0.000	0.0240	0.0000	0.0583	*
0.000 % @ 20 kHz	0.000	0.0438	0.0000	0.6083	*

DISTORTION AND NOISE TEST (AUX)

300 mV Range

0.000 % @ 10 Hz	0.000	0.1071	0.0000	0.2667	*
0.000 % @ 20 Hz	0.000	0.0154	0.0000	0.1267	*
0.000 % @ 45 Hz	0.000	0.0114	0.0000	0.1267	*
0.000 % @ 1 kHz	0.000	0.0097	0.0000	0.1467	*

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Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.000 % @ 5 kHz	0.000	0.0245	0.0000	0.3667	*
0.000 % @ 10 kHz	0.000	0.0400	0.0000	0.6667	*
3 V Range					
0.000 % @ 10 Hz	0.000	0.1082	0.0000	0.2067	*
0.000 % @ 20 Hz	0.000	0.0107	0.0000	0.0667	*
0.000 % @ 45 Hz	0.000	0.0065	0.0000	0.0667	*
0.000 % @ 1 kHz	0.000	0.0067	0.0000	0.0867	*
0.000 % @ 5 kHz	0.000	0.0087	0.0000	0.3067	*
0.000 % @ 10 kHz	0.000	0.0188	0.0000	0.6067	*

1 Year Specification Limits shown

AC VOLTAGE ACCURACY TEST (NORMAL)

33mV Range

3.000 mV @ 45 Hz	3.000	2.9994	2.9935	3.0065	1.5e-006 V
3.000 mV @ 10 kHz	3.000	2.9995	2.9935	3.0065	1.5e-006 V
30.000 mV @ 9.5 Hz	30.000	30.0165	28.3350	31.6650	3.0e-005 V
30.000 mV @ 10 Hz	30.000	29.9976	29.9700	30.0300	6.8e-006 V
30.000 mV @ 45 Hz	30.000	30.0007	29.9895	30.0105	3.0e-006 V
30.000 mV @ 1 kHz	30.000	30.0007	29.9895	30.0105	3.0e-006 V
30.000 mV @ 10 kHz	30.000	30.0012	29.9895	30.0105	3.0e-006 V
30.000 mV @ 20 kHz	30.000	30.0013	29.9880	30.0120	3.0e-006 V
30.000 mV @ 50 kHz	30.000	29.9959	29.9640	30.0360	6.1e-006 V
30.000 mV @ 100 kHz	30.000	29.9824	29.8830	30.1170	8.0e-006 V
30.000 mV @ 450 kHz	30.000	29.9565	29.7100	30.2900	2.2e-005 V

330mV Range

33.000 mV @ 45 Hz	33.000	33.0024	32.9872	33.0128	3.2e-006 V
33.000 mV @ 10 kHz	33.000	33.0022	32.9872	33.0128	3.3e-006 V
300.000 mV @ 9.5 Hz	300.000	300.0756	283.3500	316.6500	3.0e-004 V
300.000 mV @ 10 Hz	300.000	299.9676	299.9020	300.0980	5.0e-005 V
300.000 mV @ 45 Hz	300.000	299.9980	299.9485	300.0515	9.1e-006 V
300.000 mV @ 1 kHz	300.000	299.9996	299.9485	300.0515	9.1e-006 V
300.000 mV @ 10 kHz	300.000	299.9984	299.9485	300.0515	9.4e-006 V
300.000 mV @ 20 kHz	300.000	299.9928	299.9440	300.0560	9.2e-006 V
300.000 mV @ 50 kHz	300.000	299.9746	299.8870	300.1130	1.5e-005 V
300.000 mV @ 100 kHz	300.000	299.9474	299.7280	300.2720	2.5e-005 V
300.000 mV @ 500 kHz	300.000	300.0564	299.3300	300.6700	7.6e-005 V

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
3.3V Range					
0.33000 V @ 45 Hz	0.33000	0.330006	0.329891	0.330109	1.0e-005 V
0.33000 V @ 10 kHz	0.33000	0.330017	0.329891	0.330109	1.2e-005 V
3.00000 V @ 9.5 Hz	3.00000	3.000580	2.833500	3.166500	3.0e-003 V
3.00000 V @ 10 Hz	3.00000	2.999708	2.999050	3.000950	4.7e-004 V
3.00000 V @ 45 Hz	3.00000	3.000016	2.999490	3.000510	9.5e-005 V
3.00000 V @ 1 kHz	3.00000	3.000034	2.999490	3.000510	1.0e-004 V
3.00000 V @ 10 kHz	3.00000	3.000050	2.999490	3.000510	9.2e-005 V
3.00000 V @ 20 kHz	3.00000	3.000090	2.999370	3.000630	9.0e-005 V
3.00000 V @ 50 kHz	3.00000	3.000292	2.999050	3.000950	1.1e-004 V
3.00000 V @ 100 kHz	3.00000	3.000976	2.997775	3.002225	1.9e-004 V
3.00000 V @ 450 kHz	3.00000	3.000298	2.992200	3.007800	9.3e-004 V
33V Range					
3.3000 V @ 45 Hz	3.3000	3.30005	3.29891	3.30109	1.1e-004 V
3.3000 V @ 10 kHz	3.3000	3.30012	3.29891	3.30109	1.1e-004 V
30.0000 V @ 9.5 Hz	30.0000	29.98078	28.33500	31.66500	3.0e-002 V
30.0000 V @ 10 Hz	30.0000	29.99654	29.99035	30.00965	4.7e-003 V
30.0000 V @ 45 Hz	30.0000	30.00000	29.99490	30.00510	9.2e-004 V
30.0000 V @ 1 kHz	30.0000	29.99862	29.99490	30.00510	9.2e-004 V
30.0000 V @ 10 kHz	30.0000	29.99848	29.99490	30.00510	9.1e-004 V
30.0000 V @ 20 kHz	30.0000	29.99864	29.99220	30.00780	9.1e-004 V
30.0000 V @ 50 kHz	30.0000	29.99866	29.98890	30.01110	1.3e-003 V
30.0000 V @ 90 kHz	30.0000	29.99916	29.97140	30.02860	2.2e-003 V
330V Range					
33.000 V @ 45 Hz	33.000	33.0009	32.9917	33.0083	1.0e-003 V
33.000 V @ 10 kHz	33.000	33.0009	32.9874	33.0126	1.1e-003 V
100.000 V @ 50 kHz	100.000	100.0090	99.9640	100.0360	5.5e-003 V
200.000 V @ 50 kHz	200.000	200.0073	199.9340	200.0660	1.1e-002 V
200.000 V @ 100 kHz	200.000	200.0685	199.5500	200.4500	2.2e-002 V
300.000 V @ 45 Hz	300.000	300.0076	299.9410	300.0590	1.2e-002 V
300.000 V @ 1 kHz	300.000	299.9840	299.9410	300.0590	1.2e-002 V
300.000 V @ 10 kHz	300.000	299.9844	299.9340	300.0660	1.2e-002 V
300.000 V @ 18 kHz	300.000	299.9778	299.9190	300.0810	1.2e-002 V
300.000 V @ 50 kHz	300.000	299.9890	299.9040	300.0960	3.0e-002 V
1000V Range					

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
330.00 V @ 45 Hz	330.00	330.005	329.891	330.109	1.4e-002 V
330.00 V @ 10 kHz	330.00	330.006	329.891	330.109	1.6e-002 V
1000.00 V @ 45 Hz	1000.00	1000.137	999.690	1000.310	3.5e-002 V
1000.00 V @ 1 kHz	1000.00	999.994	999.740	1000.260	3.1e-002 V
1000.00 V @ 5 kHz	1000.00	999.994	999.740	1000.260	3.0e-002 V
1000.00 V @ 8 kHz	1000.00	999.977	999.690	1000.310	3.1e-002 V
1020.00 V @ 1 kHz	1020.00	1019.983	1019.735	1020.265	3.1e-002 V
1020.00 V @ 8 kHz	1020.00	1019.964	1019.684	1020.316	3.3e-002 V

AC VOLTAGE ACCURACY TEST, SQUAREWAVE (NORMAL)
66mV Range

60.000 mVpp @ 10 Hz	60.000	60.0082	59.5200	60.4800	*
60.000 mVpp @ 45 Hz	60.000	60.0036	59.6850	60.3150	*
60.000 mVpp @ 1 kHz	60.000	60.0102	59.6850	60.3150	*
60.000 mVpp @ 20 kHz	60.000	60.1460	59.5350	60.4650	*
60.000 mVpp @ 100 kHz	60.000	60.7240	56.6700	63.3300	*

660mV Range

600.000 mVpp @ 10 Hz	600.000	600.0720	595.2000	604.8000	*
600.000 mVpp @ 45 Hz	600.000	600.0200	596.8500	603.1500	*
600.000 mVpp @ 1 kHz	600.000	600.0920	596.8500	603.1500	*
600.000 mVpp @ 20 kHz	600.000	601.5940	595.3500	604.6500	*
600.000 mVpp @ 100 kHz	600.000	608.1020	566.7000	633.3000	*

6.6V Range

6.00000 Vpp @ 10 Hz	6.00000	6.000780	5.952000	6.048000	*
6.00000 Vpp @ 45 Hz	6.00000	6.000260	5.968500	6.031500	*
6.00000 Vpp @ 1 kHz	6.00000	6.000960	5.968500	6.031500	*
6.00000 Vpp @ 20 kHz	6.00000	6.015180	5.953500	6.046500	*
6.00000 Vpp @ 100 kHz	6.00000	6.076600	5.667000	6.333000	*

66V Range

60.0000 Vpp @ 10 Hz	60.0000	60.00740	59.52000	60.48000	*
60.0000 Vpp @ 45 Hz	60.0000	60.00180	59.68500	60.31500	*
60.0000 Vpp @ 1 kHz	60.0000	60.00900	59.68500	60.31500	*
60.0000 Vpp @ 20 kHz	60.0000	60.21520	59.53500	60.46500	*
60.0000 Vpp @ 100 kHz	60.0000	61.11840	56.67000	63.33000	*

AC VOLTAGE ACCURACY TEST, TRIANGLEWAVE (NORMAL)
93mV Range

Fluke Corporation	Telephone	Facsimile	Internet	Revision	2.21
1420 75th St SW, Everett WA 98203 USA	888.993.5853	425.446.6390	www.fluke.com		

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
90.000 mVpp @ 10 Hz	90.000	89.9880	89.3100	90.6900	*
90.000 mVpp @ 45 Hz	90.000	90.0036	89.5425	90.4575	*
90.000 mVpp @ 1 kHz	90.000	90.0053	89.5425	90.4575	*
90.000 mVpp @ 20 kHz	90.000	90.0116	89.3175	90.6825	*
90.000 mVpp @ 100 kHz	90.000	90.0098	85.0350	94.9650	*
930mV Range					
900.000 mVpp @ 10 Hz	900.000	899.8559	893.1000	906.9000	*
900.000 mVpp @ 45 Hz	900.000	899.9910	895.4250	904.5750	*
900.000 mVpp @ 1 kHz	900.000	900.0083	895.4250	904.5750	*
900.000 mVpp @ 20 kHz	900.000	900.0361	893.1750	906.8250	*
900.000 mVpp @ 100 kHz	900.000	900.4552	850.3500	949.6500	*
9.3V Range					
9.00000 Vpp @ 10 Hz	9.00000	8.998629	8.931000	9.069000	*
9.00000 Vpp @ 45 Hz	9.00000	9.000014	8.954250	9.045750	*
9.00000 Vpp @ 1 kHz	9.00000	9.000187	8.954250	9.045750	*
9.00000 Vpp @ 20 kHz	9.00000	9.000915	8.931750	9.068250	*
9.00000 Vpp @ 100 kHz	9.00000	9.010337	8.503500	9.496500	*
93V Range					
90.0000 Vpp @ 10 Hz	90.0000	89.98421	89.31000	90.69000	*
90.0000 Vpp @ 45 Hz	90.0000	89.99876	89.54250	90.45750	*
90.0000 Vpp @ 1 kHz	90.0000	89.99599	89.54250	90.45750	*
90.0000 Vpp @ 20 kHz	90.0000	90.01434	89.31750	90.68250	*
90.0000 Vpp @ 100 kHz	90.0000	90.11169	85.03500	94.96500	*

AC VOLTAGE ACCURACY TEST, SINEWAVE (AUX)

330mV Range

10.000 mV @ 45 Hz	10.000	10.0008	9.6200	10.3800	6.1e-006 V
10.000 mV @ 1 kHz	10.000	10.0017	9.6200	10.3800	7.2e-006 V
10.000 mV @ 5 kHz	10.000	10.0042	9.5300	10.4700	8.3e-006 V
10.000 mV @ 10 kHz	10.000	10.0047	9.5100	10.4900	9.4e-006 V
10.000 mV @ 30 kHz	10.000	10.1468	8.6000	11.4000	1.8e-005 V
300.000 mV @ 9.5 Hz	300.000	299.9882	283.3500	316.6500	3.0e-004 V
300.000 mV @ 10 Hz	300.000	299.9224	299.0300	300.9700	5.2e-005 V
300.000 mV @ 45 Hz	300.000	299.9958	299.3300	300.6700	1.7e-005 V
300.000 mV @ 1 kHz	300.000	299.9988	299.3300	300.6700	1.9e-005 V
300.000 mV @ 5 kHz	300.000	299.9938	298.9500	301.0500	2.3e-005 V

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
300.000 mV @ 10 kHz	300.000	299.9948	298.3500	301.6500	2.6e-005 V
300.000 mV @ 30 kHz	300.000	304.1808	284.1000	315.9000	5.4e-005 V
3.3V Range					
3.00000 V @ 9.5 Hz	3.00000	2.999888	2.833500	3.166500	3.0e-003 V
3.00000 V @ 10 Hz	3.00000	2.999274	2.993550	3.006450	4.8e-004 V
3.00000 V @ 45 Hz	3.00000	3.000006	2.996850	3.003150	1.8e-004 V
3.00000 V @ 1 kHz	3.00000	3.000014	2.996850	3.003150	2.2e-004 V
3.00000 V @ 5 kHz	3.00000	3.000000	2.992600	3.007400	2.5e-004 V
3.00000 V @ 10 kHz	3.00000	3.000024	2.986600	3.013400	2.8e-004 V
3.00000 V @ 30 kHz	3.00000	3.002040	2.847200	3.152800	5.3e-004 V
5V Range					
5.0000 V @ 9.5 Hz	5.0000	4.99740	4.72500	5.27500	5.0e-003 V
5.0000 V @ 10 Hz	5.0000	4.99871	4.98955	5.01045	8.0e-004 V
5.0000 V @ 45 Hz	5.0000	5.00001	4.99505	5.00495	3.0e-004 V
5.0000 V @ 1 kHz	5.0000	5.00035	4.99505	5.00495	3.6e-004 V
5.0000 V @ 5 kHz	5.0000	4.99994	4.98860	5.01140	4.7e-004 V
5.0000 V @ 10 kHz	5.0000	4.99931	4.97860	5.02140	7.0e-004 V
AC VOLTAGE ACCURACY TEST, SQUAREWAVE (AUX)					
660mV Range					
600.000 mVpp @ 10 Hz	600.000	600.0660	595.2000	604.8000	*
600.000 mVpp @ 45 Hz	600.000	600.0500	596.8500	603.1500	*
600.000 mVpp @ 1 kHz	600.000	601.0840	596.8500	603.1500	*
600.000 mVpp @ 10 kHz	600.000	611.8100	566.7000	633.3000	*
6.6V Range					
6.00000 Vpp @ 10 Hz	6.00000	6.000740	5.952000	6.048000	*
6.00000 Vpp @ 45 Hz	6.00000	6.000540	5.968500	6.031500	*
6.00000 Vpp @ 1 kHz	6.00000	6.009660	5.968500	6.031500	*
6.00000 Vpp @ 10 kHz	6.00000	6.097420	5.667000	6.333000	*
AC VOLTAGE ACCURACY TEST, TRIANGLEWAVE (AUX)					
930mV Range					
900.000 mVpp @ 10 Hz	900.000	899.6897	893.1000	906.9000	*
900.000 mVpp @ 45 Hz	900.000	899.9841	895.4250	904.5750	*
900.000 mVpp @ 1 kHz	900.000	900.0326	895.4250	904.5750	*
900.000 mVpp @ 10 kHz	900.000	900.0534	850.3500	949.6500	*

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
9.00000 Vpp @ 10 Hz	9.00000	8.997000	8.931000	9.069000	*
9.00000 Vpp @ 45 Hz	9.00000	8.999910	8.954250	9.045750	*
9.00000 Vpp @ 1 kHz	9.00000	8.999945	8.954250	9.045750	*
9.00000 Vpp @ 10 kHz	9.00000	8.983352	8.503500	9.496500	*

HARMONIC AMPLITUDE ACCURACY TEST (NORMAL)

33mV Range

30.000 mV @ 1 kHz	30.000	30.0008	29.9835	30.0165	*
30.000 mV @ 5 kHz	30.000	30.0009	29.9835	30.0165	*
30.000 mV @ 10 kHz	30.000	30.0010	29.9835	30.0165	*
30.000 mV @ 30 kHz	30.000	29.9996	29.9580	30.0420	*

330mV Range

300.000 mV @ 1 kHz	300.000	300.0010	299.9405	300.0595	*
300.000 mV @ 5 kHz	300.000	300.0000	299.9405	300.0595	*
300.000 mV @ 10 kHz	300.000	299.9980	299.9405	300.0595	*
300.000 mV @ 30 kHz	300.000	299.9860	299.8790	300.1210	*

3.3V Range

3.00000 V @ 1 kHz	3.00000	3.000050	2.999430	3.000570	*
3.00000 V @ 5 kHz	3.00000	3.000050	2.999430	3.000570	*
3.00000 V @ 10 kHz	3.00000	3.000040	2.999430	3.000570	*
3.00000 V @ 30 kHz	3.00000	3.000140	2.999000	3.001000	*

33V Range

30.0000 V @ 1 kHz	30.0000	29.99860	29.99430	30.00570	*
30.0000 V @ 5 kHz	30.0000	29.99860	29.99430	30.00570	*
30.0000 V @ 10 kHz	30.0000	29.99840	29.99430	30.00570	*
30.0000 V @ 30 kHz	30.0000	29.99860	29.98830	30.01170	*

330V Range

300.000 V @ 1 kHz	300.000	299.9850	299.9390	300.0610	*
300.000 V @ 5 kHz	300.000	299.9840	299.9280	300.0720	*
300.000 V @ 10 kHz	300.000	299.9860	299.9280	300.0720	*
300.000 V @ 30 kHz	300.000	299.9830	299.8980	300.1020	*

1000V Range

1000.00 V @ 1 kHz	1000.00	999.995	999.730	1000.270	*
1000.00 V @ 5 kHz	1000.00	999.988	999.730	1000.270	*
1000.00 V @ 10 kHz	1000.00	999.976	999.680	1000.320	*

HARMONIC AMPLITUDE ACCURACY TEST (AUX)

Fluke Corporation	Telephone	Facsimile	Internet	Revision	2.21
1420 75th St SW, Everett WA 98203 USA	888.993.5853	425.446.6390	www.fluke.com		

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
330mV Range					
300.000 mV @ 1 kHz	300.000	299.9980	299.0200	300.9800	*
300.000 mV @ 5 kHz	300.000	299.9950	298.6500	301.3500	*
300.000 mV @ 10 kHz	300.000	299.9980	298.2000	301.8000	*
300.000 mV @ 30 kHz	300.000	304.1980	286.2000	313.8000	*
3.3V Range					
3.00000 V @ 1 kHz	3.00000	3.000000	2.997000	3.003000	*
3.00000 V @ 5 kHz	3.00000	2.999990	2.992700	3.007300	*
3.00000 V @ 10 kHz	3.00000	3.000010	2.988200	3.011800	*
3.00000 V @ 30 kHz	3.00000	3.002030	2.874400	3.125600	*
5V Range					
5.00000 V @ 1 kHz	5.00000	5.000320	4.995600	5.004400	*
5.00000 V @ 5 kHz	5.00000	4.999730	4.989700	5.010300	*
5.00000 V @ 10 kHz	5.00000	4.998480	4.982200	5.017800	*
AC CURRENT ACCURACY TEST					
330 µA Range					
33.000 µA @ 1 kHz	33.000	33.0061	32.8588	33.1412	8.3e-009 A
33.000 µA @ 10 kHz	33.000	32.9925	32.5360	33.4640	2.1e-008 A
33.000 µA @ 30 kHz	33.000	32.9897	32.0720	33.9280	3.3e-008 A
190.000 µA @ 45 Hz	190.000	189.9945	189.6625	190.3375	2.0e-008 A
190.000 µA @ 1 kHz	190.000	190.0329	189.6625	190.3375	2.8e-008 A
190.000 µA @ 10 kHz	190.000	189.9083	188.2800	191.7200	7.0e-008 A
190.000 µA @ 30 kHz	190.000	189.8604	186.5600	193.4400	1.3e-007 A
329.000 µA @ 10 Hz	329.000	328.9256	328.2420	329.7580	9.1e-008 A
329.000 µA @ 45 Hz	329.000	329.0054	328.4887	329.5113	7.1e-008 A
329.000 µA @ 1 kHz	329.000	329.0710	328.4887	329.5113	4.6e-008 A
329.000 µA @ 5 kHz	329.000	329.2087	327.8630	330.1370	8.7e-008 A
329.000 µA @ 10 kHz	329.000	329.0496	326.1680	331.8320	1.5e-007 A
329.000 µA @ 30 kHz	329.000	329.0810	323.3360	334.6640	3.3e-007 A
3.3mA Range					
0.33000 mA @ 1 kHz	0.33000	0.329998	0.329520	0.330480	4.6e-008 A
0.33000 mA @ 5 kHz	0.33000	0.330003	0.329140	0.330860	8.7e-008 A
0.33000 mA @ 30 kHz	0.33000	0.329207	0.326100	0.333900	3.3e-007 A
1.90000 mA @ 1 kHz	1.90000	1.899929	1.897950	1.902050	1.2e-007 A
1.90000 mA @ 10 kHz	1.90000	1.899902	1.890200	1.909800	6.4e-007 A

Fluke Corporation

Telephone

Facsimile

Internet

Revision 2.21

1420 75th St SW, Everett WA 98203 USA

888.993.5853

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Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
1.90000 mA @ 30 kHz	1.90000	1.899755	1.880400	1.919600	1.8e-006 A
3.29000 mA @ 10 Hz	3.29000	3.289149	3.283270	3.296730	8.2e-007 A
3.29000 mA @ 45 Hz	3.29000	3.290011	3.286560	3.293440	5.3e-007 A
3.29000 mA @ 1 kHz	3.29000	3.290020	3.286560	3.293440	3.9e-007 A
3.29000 mA @ 5 kHz	3.29000	3.290086	3.283220	3.296780	4.8e-007 A
3.29000 mA @ 10 kHz	3.29000	3.290090	3.273250	3.306750	3.1e-006 A
3.29000 mA @ 30 kHz	3.29000	3.290230	3.256500	3.323500	3.2e-006 A
33mA Range					
3.3000 mA @ 1 kHz	3.3000	3.30012	3.29668	3.30332	3.9e-007 A
3.3000 mA @ 5 kHz	3.3000	3.30015	3.29536	3.30464	5.0e-007 A
3.3000 mA @ 30 kHz	3.3000	3.30447	3.28280	3.31720	2.8e-006 A
19.0000 mA @ 1 kHz	19.0000	18.99986	18.99040	19.00960	1.3e-006 A
19.0000 mA @ 10 kHz	19.0000	18.99984	18.95900	19.04100	6.7e-006 A
19.0000 mA @ 30 kHz	19.0000	19.00031	18.92000	19.08000	1.1e-005 A
32.9000 mA @ 10 Hz	32.9000	32.89441	32.83878	32.96122	4.5e-006 A
32.9000 mA @ 45 Hz	32.9000	32.90305	32.88484	32.91516	4.3e-006 A
32.9000 mA @ 1 kHz	32.9000	32.90318	32.88484	32.91516	4.5e-006 A
32.9000 mA @ 5 kHz	32.9000	32.90308	32.87168	32.92832	4.7e-006 A
32.9000 mA @ 10 kHz	32.9000	32.90387	32.83120	32.96880	8.7e-006 A
32.9000 mA @ 30 kHz	32.9000	32.90467	32.76440	33.03560	3.0e-005 A
330mA Range					
33.000 mA @ 1 kHz	33.000	33.0009	32.9668	33.0332	6.2e-006 A
33.000 mA @ 5 kHz	33.000	33.0058	32.9170	33.0830	4.7e-006 A
33.000 mA @ 30 kHz	33.000	33.0400	32.6680	33.3320	2.7e-005 A
190.000 mA @ 1 kHz	190.000	190.0017	189.9040	190.0960	1.2e-005 A
190.000 mA @ 10 kHz	190.000	190.0097	189.5200	190.4800	5.1e-005 A
190.000 mA @ 30 kHz	190.000	190.0049	189.0400	190.9600	1.2e-004 A
329.000 mA @ 10 Hz	329.000	328.9654	328.3878	329.6122	4.5e-005 A
329.000 mA @ 45 Hz	329.000	329.0955	328.8484	329.1516	4.2e-005 A
329.000 mA @ 1 kHz	329.000	329.1177	328.8484	329.1516	3.9e-005 A
329.000 mA @ 5 kHz	329.000	329.1402	328.6210	329.3790	4.8e-005 A
329.000 mA @ 10 kHz	329.000	329.1608	328.2420	329.7580	9.3e-005 A
329.000 mA @ 30 kHz	329.000	329.1689	327.4840	330.5160	2.8e-004 A
2.99A Range					
0.33000 A @ 1 kHz	0.33000	0.330039	0.329735	0.330265	5.5e-005 A

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Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.33000 A @ 5 kHz	0.33000	0.330682	0.327020	0.332980	2.0e-004 A
0.33000 A @ 10 kHz	0.33000	0.331406	0.316750	0.343250	2.2e-004 A
1.09000 A @ 10 Hz	1.09000	1.089749	1.087938	1.092062	3.1e-004 A
1.09000 A @ 45 Hz	1.09000	1.090036	1.089355	1.090645	1.2e-004 A
1.09000 A @ 1 kHz	1.09000	1.090028	1.089355	1.090645	1.2e-004 A
1.09000 A @ 5 kHz	1.09000	1.090460	1.082460	1.097540	1.8e-004 A
1.09000 A @ 10 kHz	1.09000	1.092568	1.057750	1.122250	4.4e-004 A
2.99000 A @ 10 Hz	2.99000	2.989332	2.984518	2.995482	9.7e-004 A
2.99000 A @ 45 Hz	2.99000	2.989993	2.988106	2.991894	8.8e-004 A
2.99000 A @ 1 kHz	2.99000	2.989924	2.988106	2.991894	8.5e-004 A
2.99000 A @ 5 kHz	2.99000	2.990544	2.971060	3.008940	1.1e-003 A
2.99000 A @ 10 kHz	2.99000	2.998904	2.910250	3.069750	2.0e-003 A
20.5A Range					
3.3000 A @ 500 Hz	3.3000	3.30027	3.29470	3.30530	9.3e-004 A
3.3000 A @ 1 kHz	3.3000	3.30028	3.29470	3.30530	9.3e-004 A
3.3000 A @ 5 kHz	3.3000	3.28879	3.19900	3.40100	1.2e-003 A
10.9000 A @ 45 Hz	10.9000	10.90013	10.89146	10.90854	2.6e-003 A
10.9000 A @ 65 Hz	10.9000	10.90041	10.89146	10.90854	2.6e-003 A
10.9000 A @ 500 Hz	10.9000	10.90069	10.88710	10.91290	2.6e-003 A
10.9000 A @ 1 kHz	10.9000	10.90069	10.88710	10.91290	2.6e-003 A
10.9000 A @ 5 kHz	10.9000	10.86095	10.57100	11.22900	3.6e-003 A
20.0000 A @ 45 Hz	20.0000	20.00423	19.97100	20.02900	4.6e-003 A
20.0000 A @ 65 Hz	20.0000	20.00640	19.97100	20.02900	4.6e-003 A
20.0000 A @ 500 Hz	20.0000	20.00714	19.96500	20.03500	4.6e-003 A
20.0000 A @ 1 kHz	20.0000	20.00738	19.96500	20.03500	4.6e-003 A
20.0000 A @ 5 kHz	20.0000	19.94583	19.39500	20.60500	6.4e-003 A

1 Year Specification Limits shown

RESISTANCE 2W COMP TEST

2W Comp Test 0 Ohms <= 0.002 Ohms

Pass

RESISTANCE ACCURACY TEST

11 Ohm Range

0.00000 Ohm	0.00000	-0.000016	-0.001000	0.001000	*
1.00000 Ohm	1.00000	1.000001	0.998960	1.001040	2.3e-005 Ω
2.00000 Ohm	2.00000	2.000023	1.998920	2.001080	2.9e-005 Ω
10.90000 Ohm	10.90000	10.900131	10.898564	10.901436	1.1e-004 Ω

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
33 Ohm Range					
11.9000 Ohm	11.9000	11.90016	11.89814	11.90186	8.5e-005 Ω
19.0000 Ohm	19.0000	19.00020	18.99793	19.00207	1.3e-004 Ω
30.0000 Ohm	30.0000	30.00030	29.99760	30.00240	2.0e-004 Ω
110 Ohm Range					
33.0000 Ohm	33.0000	33.00013	32.99768	33.00232	2.0e-004 Ω
109.0000 Ohm	109.0000	109.00080	108.99555	109.00445	5.5e-004 Ω
330 Ohm Range					
119.0000 Ohm	119.0000	119.00026	118.99467	119.00533	5.9e-004 Ω
190.0000 Ohm	190.0000	190.00058	189.99268	190.00732	8.9e-004 Ω
300.0000 Ohm	300.0000	300.00075	299.98960	300.01040	1.7e-003 Ω
1.1 kOhm Range					
0.330000 kOhm	0.330000	0.3300001	0.3299888	0.3300112	1.9e-003 Ω
1.090000 kOhm	1.090000	1.0900027	1.0899675	1.0900325	5.6e-003 Ω
3.3 kOhm Range					
1.190000 kOhm	1.190000	1.1899986	1.1899467	1.1900533	8.7e-003 Ω
1.900000 kOhm	1.900000	1.8999928	1.8999268	1.9000732	1.3e-002 Ω
3.000000 kOhm	3.000000	2.9999963	2.9998960	3.0001040	2.5e-002 Ω
11 kOhm Range					
3.30000 kOhm	3.30000	3.299977	3.299888	3.300112	2.6e-002 Ω
10.90000 kOhm	10.90000	10.899979	10.899675	10.900325	7.9e-002 Ω
33 kOhm Range					
11.9000 kOhm	11.9000	11.89997	11.89947	11.90053	1.2e-001 Ω
19.0000 kOhm	19.0000	18.99995	18.99927	19.00073	1.7e-001 Ω
30.0000 kOhm	30.0000	29.99997	29.99896	30.00104	2.5e-001 Ω
110 kOhm Range					
33.0000 kOhm	33.0000	32.99990	32.99888	33.00112	2.7e-001 Ω
109.0000 kOhm	109.0000	108.99996	108.99675	109.00325	8.1e-001 Ω
330 kOhm Range					
119.000 kOhm	119.000	118.9998	118.9942	119.0058	8.7e-001 Ω
190.000 kOhm	190.000	189.9998	189.9919	190.0081	1.4e+000 Ω
300.000 kOhm	300.000	299.9996	299.9884	300.0116	2.2e+000 Ω
1.1 MOhm Range					
0.330000 MOhm	0.330000	0.3299991	0.3299874	0.3300126	2.4e+000 Ω
1.090000 MOhm	1.090000	1.0900007	1.0899631	1.0900369	6.5e+000 Ω

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
3.3 MOhm Range					
1.19000 MOhm	1.19000	1.190000	1.189899	1.190101	8.8e+000 Ω
1.90000 MOhm	1.90000	1.900001	1.899856	1.900144	2.4e+001 Ω
3.00000 MOhm	3.00000	3.000002	2.999790	3.000210	3.6e+001 Ω
11 MOhm Range					
3.30000 MOhm	3.30000	3.300001	3.299521	3.300479	5.2e+001 Ω
10.90000 MOhm	10.90000	10.899738	10.898533	10.901467	2.0e+002 Ω
33 MOhm Range					
11.9000 MOhm	11.9000	11.89988	11.89453	11.90547	7.1e+002 Ω
19.0000 MOhm	19.0000	19.00039	18.99275	19.00725	9.4e+002 Ω
30.0000 MOhm	30.0000	30.00029	29.99000	30.01000	2.1e+003 Ω
110 MOhm Range					
33.0000 MOhm	33.0000	32.99912	32.98050	33.01950	3.0e+003 Ω
109.0000 MOhm	109.0000	108.99494	108.94250	109.05750	9.1e+003 Ω
330 MOhm Range					
119.000 MOhm	119.000	118.9908	118.5430	119.4570	2.2e+004 Ω
290.000 MOhm	290.000	289.9784	289.0300	290.9700	7.9e+004 Ω
1100 MOhm Range					
400.00 MOhm	400.00	399.957	393.500	406.500	4.0e+005 Ω
640.00 MOhm	640.00	639.708	629.900	650.100	6.5e+005 Ω
1090.00 MOhm	1090.00	1089.507	1073.150	1106.850	1.3e+006 Ω
CAPACITANCE ACCURACY TEST					
400 pF Range					
220.0 pF @ 1 kHz	220.0	218.43	208.90	231.10	1.7e-012 F
390.0 pF @ 1 kHz	390.0	388.68	378.05	401.95	1.6e-012 F
1.1 nF Range					
0.4800 nF @ 1 kHz	0.4800	0.47912	0.46760	0.49240	2.3e-012 F
0.6000 nF @ 1 kHz	0.6000	0.59897	0.58700	0.61300	2.6e-012 F
1.0000 nF @ 1 kHz	1.0000	0.99897	0.98500	1.01500	1.8e-012 F
3.3 nF Range					
1.2000 nF @ 1 kHz	1.2000	1.19907	1.18400	1.21600	1.8e-012 F
2.0000 nF @ 1 kHz	2.0000	1.99917	1.98000	2.02000	2.3e-012 F
3.0000 nF @ 1 kHz	3.0000	2.99930	2.97500	3.02500	3.5e-012 F
11 nF Range					
3.300 nF @ 1 kHz	3.300	3.2991	3.2818	3.3182	4.3e-012 F

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Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
7.000 nF @ 1 kHz	7.000	6.9990	6.9725	7.0275	9.5e-012 F
10.900 nF @ 1 kHz	10.900	10.8997	10.8627	10.9373	6.9e-012 F
33 nF Range					
12.000 nF @ 1 kHz	12.000	11.9990	11.8700	12.1300	7.4e-012 F
20.000 nF @ 1 kHz	20.000	19.9983	19.8500	20.1500	1.2e-011 F
30.000 nF @ 1 kHz	30.000	29.9990	29.8250	30.1750	1.8e-011 F
110 nF Range					
70.000 nF @ 1 kHz	70.000	69.9947	69.7250	70.2750	5.6e-011 F
109.000 nF @ 1 kHz	109.000	108.9933	108.6275	109.3725	6.9e-011 F
330 nF Range					
200.000 nF @ 1 kHz	200.000	199.9667	199.2000	200.8000	1.3e-010 F
300.000 nF @ 1 kHz	300.000	299.9900	298.9500	301.0500	2.2e-010 F
1.1 uF Range					
0.7000 μ F @ 100 Hz	0.7000	0.69994	0.69725	0.70275	5.7e-010 F
1.0900 μ F @ 100 Hz	1.0900	1.08993	1.08627	1.09373	7.0e-010 F
3.3 uF Range					
2.0000 μ F @ 100 Hz	2.0000	1.99980	1.99200	2.00800	1.2e-009 F
3.0000 μ F @ 100 Hz	3.0000	2.99983	2.98950	3.01050	1.9e-009 F
11 uF Range					
7.0000 μ F @ 100 Hz	7.0000	7.00040	6.97250	7.02750	5.8e-009 F
10.9000 μ F @ 100 Hz	10.9000	10.90200	10.86275	10.93725	8.8e-009 F
33 uF Range					
20.000 μ F @ 100 Hz	20.000	20.0040	19.8900	20.1100	1.6e-008 F
30.000 μ F @ 100 Hz	30.000	30.0160	29.8500	30.1500	2.8e-008 F
110 uF Range					
70.000 μ F @ 50 Hz	70.000	69.9917	69.5850	70.4150	7.8e-008 F
109.000 μ F @ 50 Hz	109.000	109.0100	108.4095	109.5905	1.3e-007 F
330 uF Range					
200.00 μ F @ 50 Hz	200.00	200.113	198.800	201.200	2.3e-007 F
300.00 μ F @ 50 Hz	300.00	300.330	298.350	301.650	3.5e-007 F
1.1 mF Range					
0.33000 mF	0.33000	0.330336	0.327515	0.332485	7.0e-007 F
0.70000 mF	0.70000	0.700432	0.695850	0.704150	*
1.09000 mF	1.09000	1.090589	1.084095	1.095905	1.2e-006 F
3.3 mF Range					

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
1.10000 mF	1.100000	1.10096	1.09205	1.10795	7.4e-007 F
2.00000 mF	2.000000	2.00144	1.98800	2.01200	1.3e-006 F
3.00000 mF	3.000000	3.00171	2.98350	3.01650	1.1e-006 F
11 mF Range					
3.3000 mF	3.30000	3.3030	3.2752	3.3248	1.7e-006 F
10.9000 mF	10.90000	10.9060	10.8409	10.9591	5.5e-006 F
33 mF Range					
20.0000 mF	20.00000	20.0144	19.8200	20.1800	1.0e-005 F
30.0000 mF	30.00000	30.0186	29.7450	30.2550	1.8e-005 F
110 mF Range					
33.000 mF	33.0000	33.038	32.537	33.463	1.7e-005 F
110.000 mF	110.0000	110.085	108.690	111.310	2.9e-005 F

THERMOCOUPLE ("J" Type) MEASUREMENT ACCURACY TEST

22.7713 °C	22.771	22.75	22.63	22.91	4.4e-002 °C
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THERMOCOUPLE SOURCING ACCURACY TEST (10uV/°C)

0.0000 mV	0.0000	-0.00011	-0.00300	0.00300	3.0e-007 V
1.0000 mV	1.0000	0.99979	0.99695	1.00305	3.1e-007 V
-1.0000 mV	-1.0000	-1.00006	-1.00305	-0.99695	3.0e-007 V
10.0000 mV	10.0000	9.99981	9.99650	10.00350	3.7e-007 V
-10.0000 mV	-10.0000	-10.00016	-10.00350	-9.99650	3.5e-007 V
100.0000 mV	100.0000	99.99918	99.99200	100.00800	1.2e-006 V
-100.0000 mV	-100.0000	-99.99955	-100.00800	-99.99200	1.2e-006 V

THERMOCOUPLE MEASUREMENT ACCURACY TEST (10uV/°C)

0.0000 mV	0.0000	0.00010	-0.00300	0.00300	4.0e-007 V
100.0000 mV	100.0000	99.99970	99.99200	100.00800	7.7e-007 V
-100.0000 mV	-100.0000	-100.00160	-100.00800	-99.99200	7.7e-007 V
300.0000 mV	300.0000	300.00170	299.98200	300.01800	9.3e-007 V
-300.0000 mV	-300.0000	-299.99680	-300.01800	-299.98200	9.3e-007 V

SC600 Option

1 Year Specification Limits shown

AMPLITUDE ACCURACY TESTS
DC VOLTAGE TEST (1 MOhm)

0.000 mV	0.000	-0.0090	-0.0400	0.0400	9.7e-006 V
1.250 mV	1.250	1.2434	1.2094	1.2906	9.5e-006 V

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
-1.250 mV	-1.250	-1.2569	-1.2906	-1.2094	9.5e-006 V
2.490 mV	2.490	2.4863	2.4488	2.5312	9.3e-006 V
-2.490 mV	-2.490	-2.4946	-2.5312	-2.4488	9.3e-006 V
2.500 mV	2.500	2.4949	2.4588	2.5412	8.9e-006 V
-2.500 mV	-2.500	-2.5054	-2.5412	-2.4588	8.9e-006 V
6.250 mV	6.250	6.2458	6.2069	6.2931	8.7e-006 V
-6.250 mV	-6.250	-6.2549	-6.2931	-6.2069	8.7e-006 V
9.900 mV	9.900	9.8963	9.8551	9.9450	8.5e-006 V
-9.900 mV	-9.900	-9.9040	-9.9450	-9.8551	8.5e-006 V
10.000 mV	10.000	9.9954	9.9550	10.0450	8.1e-006 V
-10.000 mV	-10.000	-10.0045	-10.0450	-9.9550	8.1e-006 V
17.500 mV	17.500	17.4960	17.4513	17.5487	8.9e-006 V
-17.500 mV	-17.500	-17.5044	-17.5487	-17.4513	8.9e-006 V
24.900 mV	24.900	24.8962	24.8475	24.9524	7.9e-006 V
-24.900 mV	-24.900	-24.9033	-24.9524	-24.8475	7.9e-006 V
25.000 mV	25.000	24.9946	24.9475	25.0525	7.3e-006 V
-25.000 mV	-25.000	-25.0039	-25.0525	-24.9475	7.3e-006 V
67.500 mV	67.500	67.4949	67.4262	67.5738	8.1e-006 V
-67.500 mV	-67.500	-67.5047	-67.5738	-67.4262	8.1e-006 V
109.900 mV	109.900	109.8931	109.8051	109.9950	9.5e-006 V
-109.900 mV	-109.900	-109.9013	-109.9950	-109.8051	9.5e-006 V
0.110000 V	0.110000	0.1099956	0.1099050	0.1100950	1.1e-005 V
-0.110000 V	-0.110000	-0.1099972	-0.1100950	-0.1099050	1.1e-005 V
0.305000 V	0.305000	0.3049927	0.3048075	0.3051925	1.3e-005 V
-0.305000 V	-0.305000	-0.3050004	-0.3051925	-0.3048075	1.3e-005 V
0.499000 V	0.499000	0.4989871	0.4987105	0.4992895	1.4e-005 V
-0.499000 V	-0.499000	-0.4989943	-0.4992895	-0.4987105	1.4e-005 V
0.500000 V	0.500000	0.4999873	0.4997100	0.5002900	1.8e-005 V
-0.500000 V	-0.500000	-0.4999870	-0.5002900	-0.4997100	1.8e-005 V
1.350000 V	1.350000	1.3499884	1.3492850	1.3507150	5.4e-005 V
-1.350000 V	-1.350000	-1.3500042	-1.3507150	-1.3492850	5.4e-005 V
2.19000 V	2.19000	2.189965	2.188865	2.191135	8.1e-005 V
-2.19000 V	-2.19000	-2.189969	-2.191135	-2.188865	8.1e-005 V
2.20000 V	2.20000	2.199968	2.198860	2.201140	8.1e-005 V
-2.20000 V	-2.20000	-2.199957	-2.201140	-2.198860	8.1e-005 V

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			Lower Limit	Upper Limit	
6.60000 V	6.60000	6.600039	6.596660	6.603340	2.2e-004 V
-6.60000 V	-6.60000	-6.600103	-6.603340	-6.596660	2.2e-004 V
10.99000 V	10.99000	10.989960	10.984465	10.995535	3.2e-004 V
-10.99000 V	-10.99000	-10.989951	-10.995535	-10.984465	3.2e-004 V
11.00000 V	11.00000	10.999577	10.994460	11.005540	2.7e-004 V
-11.00000 V	-11.00000	-10.999647	-11.005540	-10.994460	2.6e-004 V
70.5000 V	70.5000	70.50026	70.46471	70.53529	6.3e-004 V
-70.5000 V	-70.5000	-70.50156	-70.53529	-70.46471	6.2e-004 V
130.0000 V	130.0000	129.99942	129.93496	130.06504	9.7e-004 V
-130.0000 V	-130.0000	-130.00021	-130.06504	-129.93496	9.6e-004 V

DC VOLTAGE TEST (50 Ohm)

0.000 mV	0.000	-0.0046	-0.0400	0.0400	4.3e-006 V
2.490 mV	2.490	2.4862	2.4438	2.5362	5.7e-006 V
-2.490 mV	-2.490	-2.4922	-2.5362	-2.4438	5.7e-006 V
9.900 mV	9.900	9.8881	9.8353	9.9648	6.5e-006 V
-9.900 mV	-9.900	-9.8975	-9.9648	-9.8353	6.5e-006 V
24.900 mV	24.900	24.8767	24.7977	25.0023	8.5e-006 V
-24.900 mV	-24.900	-24.8854	-25.0023	-24.7977	8.5e-006 V
109.900 mV	109.900	109.8080	109.5853	110.2148	2.6e-005 V
-109.900 mV	-109.900	-109.8187	-110.2148	-109.5853	2.6e-005 V
499.000 mV	499.000	498.6002	497.7125	500.2875	1.7e-004 V
-499.000 mV	-499.000	-498.6086	-500.2875	-497.7125	1.7e-004 V
2.19000 V	2.19000	2.188246	2.184485	2.195515	7.4e-004 V
-2.19000 V	-2.19000	-2.188259	-2.195515	-2.184485	7.4e-004 V
6.59900 V	6.59900	6.593851	6.582463	6.615538	2.2e-003 V
-6.59900 V	-6.59900	-6.593966	-6.615538	-6.582463	2.2e-003 V

AC VOLTAGE TEST (1 MOhm)

1.000 mVpp @ 1 kHz	1.000	1.0063	0.9590	1.0410	8.1e-006 Vpp
-1.000 mVpp @ 1 kHz	-1.000	-0.9971	-1.0410	-0.9590	8.1e-006 Vpp
10.000 mVpp @ 1 kHz	10.000	10.0027	9.9500	10.0500	8.5e-006 Vpp
-10.000 mVpp @ 1 kHz	-10.000	-9.9982	-10.0500	-9.9500	8.5e-006 Vpp
25.000 mVpp @ 1 kHz	25.000	25.0067	24.9350	25.0650	9.0e-006 Vpp
-25.000 mVpp @ 1 kHz	-25.000	-24.9992	-25.0650	-24.9350	9.0e-006 Vpp
110.000 mVpp @ 1 kHz	110.000	110.0123	109.8500	110.1500	1.3e-005 Vpp
-110.000 mVpp @ 1 kHz	-110.000	-109.9997	-110.1500	-109.8500	1.3e-005 Vpp

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Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
200.00 mVpp @ 100 Hz	200.00	199.963	199.760	200.240	1.7e-005 Vpp
-200.00 mVpp @ 100 Hz	-200.00	-199.959	-200.240	-199.760	1.7e-005 Vpp
200.00 mVpp @ 1 kHz	200.00	199.962	199.760	200.240	1.8e-005 Vpp
-200.00 mVpp @ 1 kHz	-200.00	-199.962	-200.240	-199.760	1.8e-005 Vpp
200.00 mVpp @ 5 kHz	200.00	199.959	199.460	200.540	1.7e-005 Vpp
-200.00 mVpp @ 5 kHz	-200.00	-199.950	-200.540	-199.460	1.7e-005 Vpp
200.00 mVpp @ 10 kHz	200.00	199.980	199.460	200.540	2.5e-005 Vpp
-200.00 mVpp @ 10 kHz	-200.00	-199.945	-200.540	-199.460	2.4e-005 Vpp
0.50000 Vpp @ 1 kHz	0.50000	0.500026	0.499460	0.500540	3.1e-005 Vpp
-0.50000 Vpp @ 1 kHz	-0.50000	-0.500012	-0.500540	-0.499460	3.1e-005 Vpp
2.20000 Vpp @ 100 Hz	2.20000	2.200266	2.197760	2.202240	1.2e-004 Vpp
-2.20000 Vpp @ 100 Hz	-2.20000	-2.200233	-2.202240	-2.197760	1.2e-004 Vpp
2.20000 Vpp @ 1 kHz	2.20000	2.200349	2.197760	2.202240	1.3e-004 Vpp
-2.20000 Vpp @ 1 kHz	-2.20000	-2.200271	-2.202240	-2.197760	1.3e-004 Vpp
2.20000 Vpp @ 5 kHz	2.20000	2.200245	2.194460	2.205540	1.3e-004 Vpp
-2.20000 Vpp @ 5 kHz	-2.20000	-2.200170	-2.205540	-2.194460	1.3e-004 Vpp
2.2000 Vpp @ 10 kHz	2.2000	2.20037	2.19446	2.20554	1.4e-004 Vpp
-2.2000 Vpp @ 10 kHz	-2.2000	-2.20022	-2.20554	-2.19446	1.4e-004 Vpp
11.0000 Vpp @ 1 kHz	11.0000	11.00129	10.98896	11.01104	5.4e-004 Vpp
-11.0000 Vpp @ 1 kHz	-11.0000	-10.99966	-11.01104	-10.98896	5.5e-004 Vpp
130.000 Vpp @ 1 kHz	130.000	129.9529	129.8700	130.1300	1.0e-002 Vpp
-130.000 Vpp @ 1 kHz	-130.000	-129.9329	-130.1300	-129.8700	1.0e-002 Vpp

AC VOLTAGE TEST (50 Ohm)

1.000 mVpp @ 1 kHz	1.000	1.0069	0.9575	1.0425	8.0e-006 Vpp
-1.000 mVpp @ 1 kHz	-1.000	-0.9959	-1.0425	-0.9575	8.0e-006 Vpp
2.000 mVpp @ 1 kHz	2.000	2.0051	1.9550	2.0450	8.2e-006 Vpp
-2.000 mVpp @ 1 kHz	-2.000	-1.9954	-2.0450	-1.9550	8.2e-006 Vpp
5.000 mVpp @ 1 kHz	5.000	5.0005	4.9475	5.0525	8.6e-006 Vpp
-5.000 mVpp @ 1 kHz	-5.000	-4.9940	-5.0525	-4.9475	8.6e-006 Vpp
10.000 mVpp @ 1 kHz	10.000	9.9974	9.9350	10.0650	9.0e-006 Vpp
-10.000 mVpp @ 1 kHz	-10.000	-9.9894	-10.0650	-9.9350	9.0e-006 Vpp
20.000 mVpp @ 1 kHz	20.000	19.9829	19.9100	20.0900	9.2e-006 Vpp
-20.000 mVpp @ 1 kHz	-20.000	-19.9809	-20.0900	-19.9100	9.1e-006 Vpp
50.000 mVpp @ 1 kHz	50.000	49.9596	49.8350	50.1650	2.1e-005 Vpp
-50.000 mVpp @ 1 kHz	-50.000	-49.9551	-50.1650	-49.8350	2.1e-005 Vpp

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Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
100.000 mVpp @ 1 kHz	100.000	99.8992	99.7100	100.2900	3.6e-005 Vpp
-100.000 mVpp @ 1 kHz	-100.000	-99.9002	-100.2900	-99.7100	3.8e-005 Vpp
200.00 mVpp @ 1 kHz	200.00	199.817	199.460	200.540	6.8e-005 Vpp
-200.00 mVpp @ 1 kHz	-200.00	-199.808	-200.540	-199.460	6.8e-005 Vpp
0.50000 Vpp @ 1 kHz	0.50000	0.499654	0.498710	0.501290	1.7e-004 Vpp
-0.50000 Vpp @ 1 kHz	-0.50000	-0.499637	-0.501290	-0.498710	1.7e-004 Vpp
1.00000 Vpp @ 1 kHz	1.00000	0.999065	0.997460	1.002540	3.4e-004 Vpp
-1.00000 Vpp @ 1 kHz	-1.00000	-0.999056	-1.002540	-0.997460	3.4e-004 Vpp
2.00000 Vpp @ 10 Hz	2.00000	1.997868	1.994960	2.005040	6.9e-004 Vpp
-2.00000 Vpp @ 10 Hz	-2.00000	-1.997749	-2.005040	-1.994960	6.9e-004 Vpp
2.00000 Vpp @ 100 Hz	2.00000	1.997762	1.994960	2.005040	6.7e-004 Vpp
-2.00000 Vpp @ 100 Hz	-2.00000	-1.997771	-2.005040	-1.994960	6.7e-004 Vpp
2.00000 Vpp @ 1 kHz	2.00000	1.997827	1.994960	2.005040	7.1e-004 Vpp
-2.00000 Vpp @ 1 kHz	-2.00000	-1.997768	-2.005040	-1.994960	7.1e-004 Vpp
2.00000 Vpp @ 5 kHz	2.00000	1.997634	1.994960	2.005040	7.1e-004 Vpp
-2.00000 Vpp @ 5 kHz	-2.00000	-1.997775	-2.005040	-1.994960	7.1e-004 Vpp
2.00000 Vpp @ 10 kHz	2.00000	1.998546	1.994960	2.005040	7.1e-004 Vpp
-2.00000 Vpp @ 10 kHz	-2.00000	-1.997806	-2.005040	-1.994960	7.1e-004 Vpp
5.0000 Vpp @ 1 kHz	5.0000	4.99549	4.98746	5.01254	1.7e-003 Vpp
-5.0000 Vpp @ 1 kHz	-5.0000	-4.99508	-5.01254	-4.98746	1.7e-003 Vpp

AC VOLTAGE FREQUENCY TEST

10.000000000 Hz @ 2.1 V	10.0000000000	10.000000307	9.999975000	10.000025000	2.4e-006 Hz
100.00000000 Hz @ 2.1 V	100.000000000	100.00000346	99.99975000	100.00025000	1.5e-005 Hz
1.0000000 kHz @ 2.1 V	1.000000000	1.00000000	0.9999975	1.0000025	1.5e-004 Hz
10.000000 kHz @ 2.1 V	10.00000000	10.0000000	9.999975	10.000025	1.5e-003 Hz

EDGE AMPLITUDE ACCURACY TESTS

100.000 mVpp @ 1 kHz	100.000	99.7302	97.8000	102.2000	2.4e-004 Vpp
1.000 Vpp @ 1 kHz	1.000	0.9996	0.9798	1.0202	2.4e-003 Vpp
5.000 mVpp @ 10 kHz	5.000	5.0377	4.7000	5.3000	1.9e-005 Vpp
10.000 mVpp @ 10 kHz	10.000	10.0371	9.6000	10.4000	3.5e-005 Vpp
25.000 mVpp @ 10 kHz	25.000	25.1166	24.3000	25.7000	8.5e-005 Vpp
50.000 mVpp @ 10 kHz	50.000	50.0624	48.8000	51.2000	1.8e-004 Vpp
100.00 mVpp @ 10 kHz	100.00	100.134	97.800	102.200	2.8e-004 Vpp
500.00 mVpp @ 10 kHz	500.00	497.434	489.800	510.200	1.1e-003 Vpp
1.000 Vpp @ 10 kHz	1.000	0.9998	0.9798	1.0202	2.4e-003 Vpp

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
2.500 Vpp @ 10 kHz	2.500	2.5028	2.4498	2.5502	4.8e-003 Vpp
TUNNEL DIODE PULSER DRIVE AMPLITUDE					
80.0000 Vpp @ 10 kHz	80.00000	79.9050	78.4000	81.6000	4.7e-003 Vpp
WAVE GENERATOR ACCURACY TEST (SQUARE) (1 MOhm)					
1.800 mVpp @ 1000.0 Hz	1.800	1.8001	1.6460	1.9540	8.0e-006 Vpp
11.900 mVpp @ 1000.0 Hz	11.900	11.9058	11.4430	12.3570	8.9e-006 Vpp
21.900 mVpp @ 1000.0 Hz	21.900	21.9105	21.1430	22.6570	8.8e-006 Vpp
22.000 mVpp @ 1000.0 Hz	22.000	22.0054	21.2400	22.7600	9.5e-006 Vpp
56.000 mVpp @ 1000.0 Hz	56.000	56.0143	54.2200	57.7800	1.1e-005 Vpp
89.900 mVpp @ 1000.0 Hz	89.900	89.9160	87.1030	92.6970	1.2e-005 Vpp
90.00 mVpp @ 1000.0 Hz	90.00	90.021	87.200	92.800	1.4e-005 Vpp
155.00 mVpp @ 1000.0 Hz	155.00	155.022	150.250	159.750	1.7e-005 Vpp
219.00 mVpp @ 1000.0 Hz	219.00	219.015	212.330	225.670	1.9e-005 Vpp
220.00 mVpp @ 1000.0 Hz	220.00	220.017	213.300	226.700	4.4e-005 Vpp
560.00 mVpp @ 1000.0 Hz	560.00	560.069	543.100	576.900	5.1e-005 Vpp
899.00 mVpp @ 1000.0 Hz	899.00	899.016	871.930	926.070	5.7e-005 Vpp
0.9000 Vpp @ 1000.0 Hz	0.9000	0.90001	0.87290	0.92710	2.1e-004 Vpp
3.7500 Vpp @ 1000.0 Hz	3.7500	3.75021	3.63740	3.86260	2.7e-004 Vpp
6.590 Vpp @ 1000.0 Hz	6.590	6.5900	6.3922	6.7878	3.5e-004 Vpp
6.600 Vpp @ 1000.0 Hz	6.600	6.5990	6.4019	6.7981	3.0e-003 Vpp
30.800 Vpp @ 1000.0 Hz	30.800	30.7968	29.8759	31.7241	3.3e-003 Vpp
55.000 Vpp @ 1000.0 Hz	55.000	54.9923	53.3499	56.6501	4.0e-003 Vpp
55.000 Vpp @ 10.00 Hz	55.000	55.0083	53.3499	56.6501	2.9e-003 Vpp
55.000 Vpp @ 100.00 Hz	55.000	54.9851	53.3499	56.6501	2.9e-003 Vpp
55.000 Vpp @ 10.000 kHz	55.000	54.9464	53.3499	56.6501	1.3e-002 Vpp
SCOPE WAVE GENERATOR ACCURACY TEST (SINE) (1 MOhm)					
2.00 mVpp @ 1000.0 Hz	2.00	2.008	1.840	2.160	2.9e-005 Vpp
21.90 mVpp @ 1000.0 Hz	21.90	21.949	21.143	22.657	5.3e-005 Vpp
89.90 mVpp @ 1000.0 Hz	89.90	89.871	87.103	92.697	1.5e-004 Vpp
219.0 mVpp @ 1000.0 Hz	219.0	219.14	212.33	225.67	4.3e-004 Vpp
899.0 mVpp @ 1000.0 Hz	899.0	898.74	871.93	926.07	1.4e-003 Vpp
6.590 Vpp @ 1000.0 Hz	6.590	6.5878	6.3922	6.7878	1.1e-002 Vpp
55.000 Vpp @ 1000.0 Hz	55.000	54.9525	53.3499	56.6501	8.8e-002 Vpp
SCOPE WAVE GENERATOR ACCURACY TEST (TRIANGLE) (1 MOhm)					
2.50 mVpp @ 1000.0 Hz	2.50	2.512	2.325	2.675	1.3e-005 Vpp

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Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
21.90 mVpp @ 1000.0 Hz	21.90	21.864	21.143	22.657	5.1e-005 Vpp
89.90 mVpp @ 1000.0 Hz	89.90	89.802	87.103	92.697	1.5e-004 Vpp
219.0 mVpp @ 1000.0 Hz	219.0	218.24	212.33	225.67	4.5e-004 Vpp
899.0 mVpp @ 1000.0 Hz	899.0	894.80	871.93	926.07	1.5e-003 Vpp
6.590 Vpp @ 1000.0 Hz	6.590	6.5568	6.3922	6.7878	1.1e-002 Vpp
55.000 Vpp @ 1000.0 Hz	55.000	54.8753	53.3499	56.6501	9.3e-002 Vpp

SCOPE WAVE GENERATOR ACCURACY TEST (SQUARE) (50 Ohm)

1.800 mVpp @ 1000.0 Hz	1.800	1.8083	1.6460	1.9540	8.0e-006 Vpp
6.400 mVpp @ 1000.0 Hz	6.400	6.4339	6.1080	6.6920	8.1e-006 Vpp
10.900 mVpp @ 1000.0 Hz	10.900	10.9619	10.4730	11.3270	8.4e-006 Vpp
11.000 mVpp @ 1000.0 Hz	11.000	11.0304	10.5700	11.4300	8.6e-006 Vpp
28.000 mVpp @ 1000.0 Hz	28.000	28.0870	27.0600	28.9400	9.2e-006 Vpp
44.900 mVpp @ 1000.0 Hz	44.900	45.0311	43.4530	46.3470	9.9e-006 Vpp
45.00 mVpp @ 1000.0 Hz	45.00	45.256	43.550	46.450	1.1e-005 Vpp
78.00 mVpp @ 1000.0 Hz	78.00	78.449	75.560	80.440	1.2e-005 Vpp
109.00 mVpp @ 1000.0 Hz	109.00	109.616	105.630	112.370	1.3e-005 Vpp
110.00 mVpp @ 1000.0 Hz	110.00	110.634	106.600	113.400	2.1e-005 Vpp
280.00 mVpp @ 1000.0 Hz	280.00	281.577	271.500	288.500	2.7e-005 Vpp
449.00 mVpp @ 1000.0 Hz	449.00	451.530	435.430	462.570	3.4e-005 Vpp
0.450 Vpp @ 1000.0 Hz	0.450	0.4525	0.4364	0.4636	5.3e-005 Vpp
0.780 Vpp @ 1000.0 Hz	0.780	0.7843	0.7565	0.8035	6.6e-005 Vpp
1.090 Vpp @ 1000.0 Hz	1.090	1.0959	1.0572	1.1228	7.7e-005 Vpp
1.100 Vpp @ 1000.0 Hz	1.100	1.1066	1.0669	1.1331	4.9e-004 Vpp
1.800 Vpp @ 1000.0 Hz	1.800	1.8109	1.7459	1.8541	5.4e-004 Vpp
2.500 Vpp @ 1000.0 Hz	2.500	2.5148	2.4249	2.5751	4.8e-004 Vpp

SCOPE WAVE GENERATOR ACCURACY TEST (SINE) (50 Ohm)

2.00 mVpp @ 1000.0 Hz	2.00	2.029	1.840	2.160	1.2e-005 Vpp
10.90 mVpp @ 1000.0 Hz	10.90	10.964	10.473	11.327	2.3e-005 Vpp
44.90 mVpp @ 1000.0 Hz	44.90	45.065	43.453	46.347	7.5e-005 Vpp
109.00 mVpp @ 1000.0 Hz	109.00	109.564	105.630	112.370	1.8e-004 Vpp
449.0 mVpp @ 1000.0 Hz	449.0	451.35	435.43	462.57	7.3e-004 Vpp
1.090 Vpp @ 1000.0 Hz	1.090	1.0952	1.0572	1.1228	1.7e-003 Vpp
2.500 Vpp @ 1000.0 Hz	2.500	2.5176	2.4249	2.5751	5.1e-003 Vpp

SCOPE WAVE GENERATOR ACCURACY TEST (TRIANGLE) (50 Ohm)

2.50 mVpp @ 1000.0 Hz	2.50	2.528	2.325	2.675	1.7e-005 Vpp
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Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
10.90 mVpp @ 1000.0 Hz	10.90	10.926	10.473	11.327	2.5e-005 Vpp
44.90 mVpp @ 1000.0 Hz	44.90	44.881	43.453	46.347	8.7e-005 Vpp
109.0 mVpp @ 1000.0 Hz	109.0	109.23	105.63	112.37	1.8e-004 Vpp
449.0 mVpp @ 1000.0 Hz	449.0	450.77	435.43	462.57	8.5e-004 Vpp
1.090 Vpp @ 1000.0 Hz	1.090	1.0913	1.0572	1.1228	1.7e-003 Vpp
2.500 Vpp @ 1000.0 Hz	2.500	2.5138	2.4249	2.5751	5.5e-003 Vpp

TIME MARKER ACCURACY TEST

2.000000000 ns @ 1 Vp	2.0000000000	1.999999851	1.999995000	2.000005000	2.1e-016 s
5.000000000 ns @ 1 Vp	5.0000000000	4.9999996686	4.9999875000	5.0000125000	5.1e-016 s
10.000000000 ns @ 1 Vp	10.0000000000	9.999999295	9.999975000	10.000025000	1.2e-015 s
20.000000000 ns @ 1 Vp	20.0000000000	19.999998441	19.999950000	20.000050000	2.1e-015 s
50.00000000 ns @ 1 Vp	50.000000000	49.99999595	49.99987500	50.00012500	5.1e-015 s
100.00000000 ns @ 1 Vp	100.000000000	99.99999182	99.99975000	100.00025000	1.2e-014 s
10.000000000 ms @ 1 Vp	10.0000000000	9.999999162	9.999975000	10.000025000	1.2e-009 s
20.00000000 ms @ 1 Vp	20.000000000	19.99999833	19.99995000	20.00005000	2.1e-009 s
50.00000000 ms @ 1 Vp	50.000000000	50.00002285	49.99625000	50.00375000	6.1e-009 s
2.0000000000 s @ 1 Vp	2.000000000000	1.99998549368	1.99595000000	2.00405000000	9.7e-006 s
5.0000000000 s @ 1 Vp	5.000000000000	5.00005468904	4.97487500000	5.02512500000	6.0e-005 s

PULSE WIDTH VERIFICATION TESTS

2us Period

4.00 ns	4.000	4.76	1.80	6.20	4.0e-011 s
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20us Period

4.00 ns	4.000	4.50	1.80	6.20	6.6e-011 s
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200us Period

4.00 ns	4.000	4.54	1.80	6.20	8.4e-011 s
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2ms Period

40.00 ns	40.000	41.00	36.00	44.00	1.2e-010 s
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PULSE PERIOD VERIFICATION TESTS

2.0000000000 μs	2.00000000000	1.9999998704	1.9999950000	2.0000050000	2.1e-013 s
10.000000000 μs	10.0000000000	9.999999387	9.999975000	10.000025000	1.2e-012 s
20.000000000 ms	20.0000000000	19.999998590	19.999950000	20.000050000	2.0e-009 s

RESISTANCE MEASUREMENT ACCURACY TESTS

50 Ohm Range

40 Ohm	39.989	39.98	39.95	40.03	6.8e-003 Ω
50 Ohm	49.805	49.79	49.75	49.85	6.5e-003 Ω

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	

60 Ohm	60.230	60.22	60.17	60.29	9.9e-003 Ω
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1 MOhm Range

0.6 MOhm	0.60364	0.6036	0.6030	0.6042	1.3e+002 Ω
1.0 MOhm	1.00579	1.0058	1.0048	1.0068	1.3e+002 Ω
1.5 MOhm	1.50272	1.5027	1.5012	1.5042	1.4e+002 Ω

CAPACITANCE MEASUREMENT ACCURACY TEST
50pF Range

5 pF	4.233	4.66	3.52	4.94	1.8e-013 F
20 pF	19.108	19.93	17.65	20.56	3.5e-013 F
50 pF	47.898	47.92	45.00	50.79	7.0e-013 F

OVERLOAD FUNCTION VERIFICATION - DC OUTPUT

OVERLOAD VERIFICATION TEST Pass

OVERLOAD FUNCTION VERIFICATION - AC OUTPUT

OVERLOAD VERIFICATION TEST Pass

LEVELLED SINE AMPLITUDE ACCURACY TEST (50 Ohm)

5.000 mVpp @ 50.00 kHz	5.000	5.0073	4.6000	5.4000	7.6e-006 Vpp
10.000 mVpp @ 50.00 kHz	10.000	9.9619	9.5000	10.5000	7.5e-006 Vpp
20.000 mVpp @ 50.00 kHz	20.000	20.0696	19.3000	20.7000	7.7e-006 Vpp
50.00 mVpp @ 50.00 kHz	50.00	50.176	48.700	51.300	1.3e-005 Vpp
100.00 mVpp @ 50.00 kHz	100.00	99.606	97.700	102.300	1.5e-005 Vpp
200.00 mVpp @ 50.00 kHz	200.00	200.778	195.700	204.300	1.7e-005 Vpp
0.50000 Vpp @ 50.00 kHz	0.50000	0.501628	0.489700	0.510300	3.5e-005 Vpp
1.00000 Vpp @ 50.00 kHz	1.00000	1.001147	0.979700	1.020300	5.6e-005 Vpp
2.00000 Vpp @ 50.00 kHz	2.00000	2.010149	1.959700	2.040300	9.6e-005 Vpp
5.0000 Vpp @ 50.00 kHz	5.0000	4.99959	4.89970	5.10030	3.2e-004 Vpp

LEVELLED SINE FLATNESS TEST (relative to 50kHz)
5.5V Range
5.5V Applied

0.00 % @ 1.00 MHz	0.000	-0.05	-1.50	1.50	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.03	-1.50	1.50	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.16	-1.50	1.50	8.5e-001 %
0.00 % @ 100.00 MHz	0.000	-0.03	-1.50	1.50	8.4e-001 %
0.00 % @ 200.00 MHz	0.000	-0.76	-2.00	2.00	8.8e-001 %
0.00 % @ 300.00 MHz	0.000	-0.37	-2.00	2.00	8.6e-001 %
0.00 % @ 400.00 MHz	0.000	-0.42	-4.00	4.00	8.5e-001 %

Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.00 % @ 500.00 MHz	0.000	-0.59	-4.00	4.00	8.6e-001 %
0.00 % @ 550.00 MHz	0.000	-0.27	-4.00	4.00	9.2e-001 %
0.00 % @ 600.00 MHz	0.000	-0.33	-4.00	4.00	9.4e-001 %
3.4V Applied					
0.00 % @ 1.00 MHz	0.000	-0.04	-1.50	1.50	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.08	-1.50	1.50	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.40	-1.50	1.50	8.5e-001 %
0.00 % @ 100.00 MHz	0.000	-0.54	-1.50	1.50	8.4e-001 %
0.00 % @ 200.00 MHz	0.000	-0.84	-2.00	2.00	8.8e-001 %
0.00 % @ 300.00 MHz	0.000	-0.33	-2.00	2.00	8.6e-001 %
0.00 % @ 400.00 MHz	0.000	-0.08	-4.00	4.00	8.5e-001 %
0.00 % @ 500.00 MHz	0.000	-0.28	-4.00	4.00	8.6e-001 %
0.00 % @ 550.00 MHz	0.000	-0.03	-4.00	4.00	9.2e-001 %
0.00 % @ 600.00 MHz	0.000	-0.36	-4.00	4.00	9.4e-001 %
1.3V Applied					
0.00 % @ 1.00 MHz	0.000	-0.03	-1.51	1.51	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.08	-1.51	1.51	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.48	-1.51	1.51	8.5e-001 %
0.00 % @ 100.00 MHz	0.000	-0.67	-1.51	1.51	8.4e-001 %
0.00 % @ 200.00 MHz	0.000	-0.91	-2.01	2.01	8.8e-001 %
0.00 % @ 300.00 MHz	0.000	-0.36	-2.01	2.01	8.6e-001 %
0.00 % @ 400.00 MHz	0.000	1.35	-4.01	4.01	8.5e-001 %
0.00 % @ 500.00 MHz	0.000	1.14	-4.01	4.01	8.6e-001 %
0.00 % @ 550.00 MHz	0.000	1.37	-4.01	4.01	9.2e-001 %
0.00 % @ 600.00 MHz	0.000	-0.44	-4.01	4.01	9.4e-001 %
1.3V Range					
1.2V Applied					
0.00 % @ 1.00 MHz	0.000	-0.02	-1.51	1.51	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.04	-1.51	1.51	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.42	-1.51	1.51	8.3e-001 %
0.00 % @ 100.00 MHz	0.000	-0.54	-1.51	1.51	8.2e-001 %
0.00 % @ 200.00 MHz	0.000	-0.61	-2.01	2.01	8.8e-001 %
0.00 % @ 300.00 MHz	0.000	-0.71	-2.01	2.01	8.6e-001 %
0.00 % @ 400.00 MHz	0.000	-0.37	-4.01	4.01	8.5e-001 %
0.00 % @ 500.00 MHz	0.000	-0.88	-4.01	4.01	8.6e-001 %

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Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.00 % @ 550.00 MHz	0.000	-0.29	-4.01	4.01	9.2e-001 %
0.00 % @ 600.00 MHz	0.000	-0.62	-4.01	4.01	9.4e-001 %
400mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.02	-1.53	1.53	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.03	-1.53	1.53	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.44	-1.53	1.53	8.3e-001 %
0.00 % @ 100.00 MHz	0.000	-0.53	-1.53	1.53	8.2e-001 %
0.00 % @ 200.00 MHz	0.000	-0.54	-2.03	2.03	8.8e-001 %
0.00 % @ 300.00 MHz	0.000	-0.57	-2.03	2.03	8.6e-001 %
0.00 % @ 400.00 MHz	0.000	1.84	-4.03	4.03	8.5e-001 %
0.00 % @ 500.00 MHz	0.000	1.35	-4.03	4.03	8.6e-001 %
0.00 % @ 550.00 MHz	0.000	1.91	-4.03	4.03	9.2e-001 %
0.00 % @ 600.00 MHz	0.000	-0.54	-4.03	4.03	9.4e-001 %
400mV Range					
340mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.01	-1.53	1.53	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.04	-1.53	1.53	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.40	-1.53	1.53	8.3e-001 %
0.00 % @ 100.00 MHz	0.000	-0.47	-1.53	1.53	8.1e-001 %
0.00 % @ 200.00 MHz	0.000	-0.55	-2.03	2.03	8.2e-001 %
0.00 % @ 300.00 MHz	0.000	-0.73	-2.03	2.03	8.3e-001 %
0.00 % @ 400.00 MHz	0.000	-0.33	-4.03	4.03	8.3e-001 %
0.00 % @ 500.00 MHz	0.000	-0.46	-4.03	4.03	8.4e-001 %
0.00 % @ 550.00 MHz	0.000	-0.36	-4.03	4.03	8.5e-001 %
0.00 % @ 600.00 MHz	0.000	-0.27	-4.03	4.03	8.6e-001 %
100mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.02	-1.60	1.60	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.05	-1.60	1.60	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.33	-1.60	1.60	8.3e-001 %
0.00 % @ 100.00 MHz	0.000	-0.46	-1.60	1.60	8.2e-001 %
0.00 % @ 200.00 MHz	0.000	-0.42	-2.10	2.10	8.2e-001 %
0.00 % @ 300.00 MHz	0.000	-0.52	-2.10	2.10	8.3e-001 %
0.00 % @ 400.00 MHz	0.000	2.09	-4.10	4.10	8.3e-001 %
0.00 % @ 500.00 MHz	0.000	1.91	-4.10	4.10	8.4e-001 %
0.00 % @ 550.00 MHz	0.000	1.99	-4.10	4.10	8.5e-001 %

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.00 % @ 600.00 MHz	0.000	-0.15	-4.10	4.10	8.6e-001 %
100mV Range					
99mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.02	-1.60	1.60	4.6e-001 %
0.00 % @ 10.00 MHz	0.000	-0.06	-1.60	1.60	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.36	-1.60	1.60	8.3e-001 %
0.00 % @ 100.00 MHz	0.000	-0.44	-1.60	1.60	8.2e-001 %
0.00 % @ 200.00 MHz	0.000	-0.42	-2.10	2.10	8.5e-001 %
0.00 % @ 300.00 MHz	0.000	-0.50	-2.10	2.10	8.7e-001 %
0.00 % @ 400.00 MHz	0.000	-0.07	-4.10	4.10	8.9e-001 %
0.00 % @ 500.00 MHz	0.000	-0.66	-4.10	4.10	9.1e-001 %
0.00 % @ 550.00 MHz	0.000	-0.03	-4.10	4.10	9.3e-001 %
0.00 % @ 600.00 MHz	0.000	-0.89	-4.10	4.10	9.4e-001 %
40mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.03	-1.75	1.75	4.7e-001 %
0.00 % @ 10.00 MHz	0.000	-0.03	-1.75	1.75	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.07	-1.75	1.75	8.8e-001 %
0.00 % @ 100.00 MHz	0.000	-0.24	-1.75	1.75	8.8e-001 %
0.00 % @ 200.00 MHz	0.000	-0.18	-2.25	2.25	8.9e-001 %
0.00 % @ 300.00 MHz	0.000	-0.06	-2.25	2.25	9.0e-001 %
0.00 % @ 400.00 MHz	0.000	2.35	-4.25	4.25	9.1e-001 %
0.00 % @ 500.00 MHz	0.000	1.91	-4.25	4.25	9.3e-001 %
0.00 % @ 550.00 MHz	0.000	2.34	-4.25	4.25	9.4e-001 %
0.00 % @ 600.00 MHz	0.000	0.15	-4.25	4.25	9.6e-001 %
40mV Range					
34mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.03	-1.79	1.79	4.7e-001 %
0.00 % @ 10.00 MHz	0.000	-0.02	-1.79	1.79	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.14	-1.79	1.79	8.8e-001 %
0.00 % @ 100.00 MHz	0.000	-0.32	-1.79	1.79	8.8e-001 %
0.00 % @ 200.00 MHz	0.000	-0.35	-2.29	2.29	8.9e-001 %
0.00 % @ 300.00 MHz	0.000	-0.39	-2.29	2.29	9.0e-001 %
0.00 % @ 400.00 MHz	0.000	-0.08	-4.29	4.29	9.1e-001 %
0.00 % @ 500.00 MHz	0.000	-0.20	-4.29	4.29	9.3e-001 %
0.00 % @ 550.00 MHz	0.000	-0.07	-4.29	4.29	9.4e-001 %

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
0.00 % @ 600.00 MHz	0.000	0.11	-4.29	4.29	9.6e-001 %
10mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.03	-2.50	2.50	4.7e-001 %
0.00 % @ 10.00 MHz	0.000	-0.03	-2.50	2.50	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.09	-2.50	2.50	8.9e-001 %
0.00 % @ 100.00 MHz	0.000	-0.27	-2.50	2.50	8.9e-001 %
0.00 % @ 200.00 MHz	0.000	-0.17	-3.00	3.00	9.0e-001 %
0.00 % @ 300.00 MHz	0.000	-0.14	-3.00	3.00	9.1e-001 %
0.00 % @ 400.00 MHz	0.000	2.30	-5.00	5.00	9.3e-001 %
0.00 % @ 500.00 MHz	0.000	2.20	-5.00	5.00	9.4e-001 %
0.00 % @ 550.00 MHz	0.000	2.28	-5.00	5.00	9.6e-001 %
0.00 % @ 600.00 MHz	0.000	0.23	-5.00	5.00	9.8e-001 %
10mV Range					
9.9mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.04	-2.51	2.51	4.7e-001 %
0.00 % @ 10.00 MHz	0.000	-0.03	-2.51	2.51	4.7e-001 %
0.00 % @ 50.00 MHz	0.000	-0.17	-2.51	2.51	8.9e-001 %
0.00 % @ 100.00 MHz	0.000	-0.36	-2.51	2.51	8.9e-001 %
0.00 % @ 200.00 MHz	0.000	-0.34	-3.01	3.01	9.0e-001 %
0.00 % @ 300.00 MHz	0.000	-0.32	-3.01	3.01	9.2e-001 %
0.00 % @ 400.00 MHz	0.000	0.07	-5.01	5.01	9.3e-001 %
0.00 % @ 500.00 MHz	0.000	-0.15	-5.01	5.01	9.4e-001 %
0.00 % @ 550.00 MHz	0.000	-0.06	-5.01	5.01	9.6e-001 %
0.00 % @ 600.00 MHz	0.000	0.08	-5.01	5.01	9.8e-001 %
5mV Applied					
0.00 % @ 1.00 MHz	0.000	-0.06	-3.50	3.50	4.7e-001 %
0.00 % @ 10.00 MHz	0.000	0.01	-3.50	3.50	5.0e-001 %
0.00 % @ 50.00 MHz	0.000	-0.24	-3.50	3.50	1.1e+000 %
0.00 % @ 100.00 MHz	0.000	-0.35	-3.50	3.50	1.1e+000 %
0.00 % @ 200.00 MHz	0.000	-0.26	-4.00	4.00	1.1e+000 %
0.00 % @ 300.00 MHz	0.000	-0.26	-4.00	4.00	1.1e+000 %
0.00 % @ 400.00 MHz	0.000	1.33	-6.00	6.00	1.1e+000 %
0.00 % @ 500.00 MHz	0.000	1.11	-6.00	6.00	1.1e+000 %
0.00 % @ 550.00 MHz	0.000	1.19	-6.00	6.00	1.1e+000 %
0.00 % @ 600.00 MHz	0.000	0.08	-6.00	6.00	1.2e+000 %

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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
SCOPE EDGE FREQUENCY VERIFICATION TEST					
1.000000000 kHz @ 2.5 Vpp	1.0000000000	0.999999870	0.999997500	1.000002500	1.7e-005 Hz
10.000000000 kHz @ 2.5 Vpp	10.0000000000	9.99999997	9.99997500	10.00002500	1.2e-004 Hz
100.000000000 kHz @ 2.5 Vpp	100.0000000000	100.0000056	99.9997500	100.0002500	4.2e-004 Hz
1.0000000000 MHz @ 2.5 Vpp	1.00000000000	1.0000000712	0.9999975000	1.0000025000	1.5e-003 Hz
10.0000000000 MHz @ 2.5 Vpp	10.00000000000	10.000000768	9.999975000	10.000025000	7.1e-003 Hz
EDGE RISE TIME TEST					
250 mVpp					
300.0 ps @ 1 kHz	300.00	235.4	200.0	300.0	1.2e-011 s
300.0 ps @ 1 MHz	300.00	232.2	200.0	300.0	1.3e-011 s
350.0 ps @ 10 MHz	350.00	249.1	0.0	350.0	1.2e-011 s
500 mVpp					
300.0 ps @ 1 kHz	300.00	246.6	200.0	300.0	1.4e-011 s
300.0 ps @ 1 MHz	300.00	236.8	200.0	300.0	1.2e-011 s
350.0 ps @ 10 MHz	350.00	238.3	0.0	350.0	1.2e-011 s
1 Vpp					
300.0 ps @ 1 kHz	300.00	250.3	200.0	300.0	1.3e-011 s
300.0 ps @ 1 MHz	300.00	238.4	200.0	300.0	1.2e-011 s
350.0 ps @ 10 MHz	350.00	238.3	0.0	350.0	1.2e-011 s
2.5 Vpp					
300.0 ps @ 1 kHz	300.00	232.1	200.0	300.0	1.3e-011 s
300.0 ps @ 1 MHz	300.00	227.6	200.0	300.0	1.2e-011 s
350.0 ps @ 10 MHz	350.00	243.2	0.0	350.0	1.2e-011 s
Aberration Verification		Pass			
LEVELLED SINE FREQUENCY TEST					
50.000000 kHz @ 5.5 V	50.00000000	49.999984	49.999875	50.000125	1.0e-002 Hz
500.000000 kHz @ 5.5 V	500.00000000	499.999980	499.998750	500.001250	8.9e-003 Hz
5.000000000 MHz @ 5.5 V	5.0000000000	4.999999823	4.999987500	5.000012500	1.1e-002 Hz
50.000000000 MHz @ 5.5 V	50.0000000000	49.99999824	49.99987500	50.00012500	5.7e-002 Hz
500.0000000 MHz @ 5.5 V	500.00000000	499.9999799	499.9987500	500.0012500	6.2e-001 Hz
LEVELLED SINE HARMONICS TEST					
Fundamental Frequency/Level : (50 kHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-39	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-50	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-71	-200	-38	*
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Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
-100 dB @ 5 th_harmonic	-100.0	-74	-200	-38	*
Fundamental Frequency/Level : (100 kHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-47	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-51	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-72	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-76	-200	-38	*
Fundamental Frequency/Level : (200 kHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-50	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-52	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-71	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-77	-200	-38	*
Fundamental Frequency/Level : (400 kHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-46	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-44	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-59	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-73	-200	-38	*
Fundamental Frequency/Level : (800 kHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-52	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-51	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-74	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-78	-200	-38	*
Fundamental Frequency/Level : (1 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-78	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-51	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-75	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-78	-200	-38	*
Fundamental Frequency/Level : (2 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-85	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-86	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-80	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-81	-200	-38	*
Fundamental Frequency/Level : (4 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-47	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-51	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-74	-200	-38	*

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Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
-100 dB @ 5 th_harmonic	-100.0	-80	-200	-38	*
Fundamental Frequency/Level : (8 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-43	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-50	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-71	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-79	-200	-38	*
Fundamental Frequency/Level : (10 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-42	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-49	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-71	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-78	-200	-38	*
Fundamental Frequency/Level : (20 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-39	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-48	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-71	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-79	-200	-38	*
Fundamental Frequency/Level : (40 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-38	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-47	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-66	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-76	-200	-38	*
Fundamental Frequency/Level : (80 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-40	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-43	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-58	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-69	-200	-38	*
Fundamental Frequency/Level : (100 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-43	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-43	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-61	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-66	-200	-38	*
Fundamental Frequency/Level : (200 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-45	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-53	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-78	-200	-38	*

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Certificate Number: EVL918262

Date of Calibration: 4-Oct-2023

Calibration Data

Parameter	Nominal Value	Measurement Result	Limits of Error		Expanded Uncertainty
			Lower Limit	Upper Limit	
-100 dB @ 5 th_harmonic	-100.0	-81	-200	-38	*
Fundamental Frequency/Level : (400 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-43	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-70	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-81	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-80	-200	-38	*
Fundamental Frequency/Level : (600 MHz @ 5.5 Vpp)					
-100 dB @ 2 nd_harmonic	-100.0	-57	-200	-33	*
-100 dB @ 3 rd_harmonic	-100.0	-73	-200	-38	*
-100 dB @ 4 th_harmonic	-100.0	-79	-200	-38	*
-100 dB @ 5 th_harmonic	-100.0	-79	-200	-38	*